

Basic New Keynesian Model

Reference

Galí, Jordi (2008). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*, Princeton University Press.

◆ Dynamic Stochastic General Equilibrium Model

◆ Monopolistic Competition

◆ Price Rigidity

◆ Short-run Non-neutrality of Monetary Policy

Government

Budget constraint

$$T_t = M_t - M_{t-1}$$

where T_t are lump-sum transfers and M_t is the quantity of money at the end of period t .

Policy Rule

1. Money Growth Rule
2. Taylor Rule

We'll specify these policy rules later.

Firms

There's a continuum of firms indexed by $i \in [0, 1]$. Each firm produces a differentiated product, but they all use an identical technology, represented by the production function

$$Y_t(i) = A_t N_t(i)^{1-\alpha} \quad \alpha \in (0, 1)$$

where A_t evolves exogenously according to

$$\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$$

with $A > 0$, $\rho_a \in (0, 1)$, and $\mathbb{E}\{\varepsilon_{t+1}^A\} = 0$.

The cost function associated to the production function above is:

$$\Psi_t(Y_t(i)) = W_t \underbrace{A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{1}{1-\alpha}}}_{N_t}$$

Marginal cost is:

$$\psi_t(i) \equiv \Psi'_t(Y_t(i)) = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{\alpha}{1-\alpha}}$$

Monopolistic Competition: Each firm operates in a monopolistically competitive market and faces a demand schedule to be derived below.

Price Rigidity [Calvo (1983), Yun (1996)]: In any period t , each firm may reset its price with probability $1 - \theta$, independently of the time elapsed since the last adjustment. Thus, in each period, a measure $1 - \theta$ of producers reset their prices, while a fraction θ keep their prices unchanged. With this formulation, the expected duration of a price is $\frac{1}{1-\theta}$ (see below). Therefore, θ becomes a natural index of price stickiness ($\theta \rightarrow 0$ full flexibility).

The probability that a price is fixed for $k \in \{1, 2, \dots\}$ periods is

$$\Pr(k) = \theta^{k-1} (1 - \theta).$$

Notice that $\sum_{k=1}^{\infty} \Pr(k) = 1$.

Then, expected duration (ED) is:

$$\begin{aligned}
 ED &= \sum_{k=1}^{\infty} \text{Pr}(k)k \\
 &= \sum_{k=1}^{\infty} \theta^{k-1} (1 - \theta) k \\
 &= (1 - \theta) \sum_{k=1}^{\infty} k \theta^{k-1}
 \end{aligned}$$

Then:

$$\begin{aligned}
 ED &= (1 - \theta) [1 + 2\theta + 3\theta^2 + 4\theta^3 + \dots] \\
 &= (1 - \theta) [1 + \theta + \theta + \theta^2 + \theta^2 + \theta^2 + \theta^3 + \theta^3 + \theta^3 + \theta^3 + \dots] \\
 &= (1 - \theta) [(1 + \theta + \theta^2 + \theta^3 + \dots) + (\theta + \theta^2 + \theta^3 + \dots) + (\theta^2 + \theta^3 + \theta^4 + \dots) + \dots] \\
 &= (1 - \theta) [(1 + \theta + \theta^2 + \theta^3 + \dots) + \theta(1 + \theta + \theta^2 + \dots) + \theta^2(1 + \theta + \theta^2 + \dots) + \dots] \\
 &= (1 - \theta) \left[\frac{1}{1 - \theta} + \theta \frac{1}{1 - \theta} + \theta^2 \frac{1}{1 - \theta} + \dots \right] \\
 &= (1 - \theta) \frac{1}{1 - \theta} [1 + \theta + \theta^2 + \dots] \\
 &= [1 + \theta + \theta^2 + \dots] \\
 &= \frac{1}{1 - \theta}
 \end{aligned}$$

Representative Household

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U \left(C_t, N_t, \frac{M_t}{P_t} \right)$$

$$U \left(C_t, N_t, \frac{M_t}{P_t} \right) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} + \frac{\chi}{1-\nu} \left(\frac{M_t}{P_t} \right)^{1-\nu}$$

$$C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

with $\sigma > 0$, $\varphi > 0$, $\nu > 0$, $\epsilon > 1$, $\chi > 0$, and

$$\int_0^1 P_t(i) C_t(i) di + B_t + M_t = W_t N_t + (1+i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t$$

where Ω_t = nominal profits at t , and T_t = lump-sum transfers at t .

Remark. Setting $\chi = 0$ we get a model without money in the utility function. When $\sigma = 1$ replace $\frac{C_t^{1-\sigma}}{1-\sigma}$ with $\ln C$. Analogously for $\nu = 1$.

Utility Maximization

We split the problem in two: an intratemporal problem and an intertemporal problem.

Intratemporal Problem

For any given level of expenditure in consumption goods, I , maximize the value of the consumption index C_t :

$$\begin{aligned} \max_{C_t(\cdot)} \quad & \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \\ \text{s.t.} \quad & \int_0^1 P_t(i) C_t(i) di = I \end{aligned}$$

Lagrangian

$$\mathcal{L} = \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P_t(i) C_t(i) di \right]$$

FOC (see the Appendix). For each $h \in [0, 1]$ we have:

$$\frac{\epsilon}{\epsilon-1} \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}-1} \frac{\epsilon-1}{\epsilon} C_t(h)^{\frac{\epsilon-1}{\epsilon}-1} - \lambda P_t(h) = 0$$

Then:

$$\left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C_t(h)^{\frac{-1}{\epsilon}} = \lambda P_t(h) \quad \forall h \in [0, 1]$$

Then, for each pair of goods, j and h , we get:

$$\left(\frac{C_t(h)}{C_t(j)} \right)^{-\frac{1}{\epsilon}} = \frac{P_t(h)}{P_t(j)}$$

Notice that ϵ measures the elasticity of substitution between goods: $-\frac{d \ln(\frac{C_t(h)}{C_t(j)})}{d \ln(\frac{P_t(h)}{P_t(j)})} = \epsilon$.

From the expression above we get (changing index from h to i):

$$C_t(i) = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j) \quad \forall i \in [0, 1]$$

Substituting into the constraint:

$$I = \int_0^1 P_t(i) C_t(i) di = \int_0^1 P_t(i) \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j) di = \frac{C_t(j)}{P_t(j)^{-\epsilon}} \int_0^1 P_t(i)^{1-\epsilon} di$$

Then:

$$C_t(j) = P_t(j)^{-\epsilon} \frac{I}{\int_0^1 P_t(i)^{1-\epsilon} di}$$

Now let the price level be:

$$P_t = \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}}$$

Remark. P_t is defined by:

$$P_t \equiv \min_{C_t(\cdot)} \int_0^1 P_t(i) C_t(i) di$$

$$\text{s.t.} \quad C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} = 1$$

Then, our previous expression for $C_t(j)$ becomes: $C_t(j) = P_t(j)^{-\epsilon} \frac{I}{P_t^{1-\epsilon}}$. Rearranging, and changing the index to i , we can write:

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} \frac{I}{P_t}$$

Substituting the expression above into the objective we get:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} = \left(\int_0^1 \left[\left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} \frac{I}{P_t} \right]^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

$$C_t = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} \left(\int_0^1 P_t(i)^{1-\epsilon} di \right)^{-\frac{\epsilon}{1-\epsilon}} = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} \left(P_t^{1-\epsilon} \right)^{-\frac{\epsilon}{1-\epsilon}} = \frac{I}{P_t} \frac{1}{P_t^{-\epsilon}} P_t^{-\epsilon}$$

Then:

$$C_t = \frac{I}{P_t}$$

Substituting the expression above into the one for $C_t(i)$ we get:

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} C_t \quad \forall i \in [0, 1]$$

Also, notice that $C_t = \frac{I}{P_t}$ implies $P_t C_t = I$, and then:

$$P_t C_t = \int_0^1 P_t(i) C_t(i) di$$

Intertemporal Problem

Using our last result we can write the intertemporal problem as follows:

$$\begin{aligned} \max \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ \text{s.t.} \quad & P_t C_t + B_t + M_t = W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t \end{aligned}$$

Lagrangian

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right) + \right. \\ & \left. \sum_{t=0}^{\infty} \beta^t \lambda_t \left[W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} \right. \right. \\ & \left. \left. + \Omega_t + T_t - P_t C_t - B_t - M_t \right] \right\} \end{aligned}$$

FOC

$$C_t : \quad U_{c,t} = \lambda_t P_t$$

$$N_t : \quad -U_{n,t} = \lambda_t W_t$$

$$M_t : \quad \lambda_t = U_{m,t} \frac{1}{P_t} + \beta \mathbb{E}_t \{ \lambda_{t+1} \}$$

$$B_t : \quad \lambda_t = \beta (1 + i_t) \mathbb{E}_t \{ \lambda_{t+1} \}$$

Remark. $U_m \equiv \frac{\partial U}{\partial (\frac{M}{P})}$.

Combining the FOCs w.r.t. C_t and N_t we get:

$$\frac{-U_{n,t}}{U_{c,t}} = \frac{W_t}{P_t}$$

Combining the FOCs w.r.t. C_t and B_t we get:

$$U_{c,t} = (1 + i_t) \beta \mathbb{E}_t \left\{ U_{c,t+1} \frac{P_t}{P_{t+1}} \right\}$$

Combining the FOCs w.r.t. C_t and M_t we get:

$$U_{c,t} = U_{m,t} + \beta \mathbb{E}_t \left\{ U_{c,t+1} \frac{P_t}{P_{t+1}} \right\}$$

Combining the previous two expressions we get:

$$\frac{U_{m,t}}{U_{c,t}} = \frac{i_t}{1 + i_t}$$

For our particular utility function we have:

$$\begin{aligned} U_{c,t} &= C_t^{-\sigma} \\ U_{n,t} &= -N_t^\varphi \\ U_{m,t} &= \chi \left(\frac{M_t}{P_t} \right)^{-\nu} \end{aligned}$$

Then, our previous expressions become ($d = \text{demand}$):

$$C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$$

$$\frac{M_t^d}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1 + i_t} \right)^{-\frac{1}{\nu}}$$

$$1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$$

Stochastic Discount Factor

The stochastic discount factor for one-period nominal payoffs is:

$$Q_{t,t+1} = \beta \frac{U_{c,t+1}}{U_{c,t}} \frac{P_t}{P_{t+1}} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}}$$

Analogously, for k -period nominal payoffs we have:

$$Q_{t,t+k} = \beta^k \frac{U_{c,t+k}}{U_{c,t}} \frac{P_t}{P_{t+k}} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}}$$

Intuition: Think of buying, in period t , S_t nominal units of an asset that will pay the (possibly random) amount D_{t+k} per unit in period $t+k$. Let q_t be the per-unit price of this asset. Then, you spend $q_t S_t$ in period t in order to get $S_t D_{t+k}$ (possibly random) nominal units in period $t+k$. Then, combining the FOC w.r.t. C_t and S_t we can show that q_t will satisfy $q_t = \mathbb{E}_t \left\{ \beta^k \frac{U_{c,t+k}}{U_{c,t}} \frac{P_t}{P_{t+k}} D_{t+k} \right\}$. That is: $q_t = \mathbb{E}_t \left\{ Q_{t,t+k} D_{t+k} \right\}$.

Optimal Price Setting by Firms

If firm $i \in [0, 1]$ gets to reoptimize in period t , it will choose the price $P_t^*(i)$ that maximizes the current market value of the profits generated while that price remains effective. The probability that the price set at t remains fixed at $t + k$, $P_{t+k}(i) = P_t^*(i)$, is θ^k for $k \in \{0, 1, 2, \dots\}$. Let $Y_{t+k|t}(i)$ denote output in period $t + k$ for a firm that last reset its price in period t . Then, nominal profits in $t + k$ will be: $\Omega_{t+k|t}(i) = P_t^*(i)Y_{t+k|t}(i) - \Psi_{t+k}(Y_{t+k|t}(i))$. Since the firm is owned by the household, and nominal profits are stochastic, the appropriate discount factor is $Q_{t,t+k}$. Hence, the optimal $P_t^*(i)$ solves:

$$\begin{aligned} \max \quad & \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[P_t^*(i) Y_{t+k|t}(i) - \Psi_{t+k}(Y_{t+k|t}(i)) \right] \right\} \\ \text{s.t.} \quad & Y_{t+k|t}(i) = \left(\frac{P_t^*(i)}{P_{t+k}} \right)^{-\epsilon} C_{t+k} \end{aligned}$$

where the processes for P_{t+k} , C_{t+k} , and then $Q_{t,t+k}$, are taken as given.

Notice that the monopolistically competitive producer is constrained by the demand of its product that was derived from utility maximization.

From the structure of the problem above we see that all firms setting their prices at t will choose the same price. Hence, from now on we drop the i index.

Substituting the constraint into the objective we get:

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[\frac{P_t^{*1-\epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} - \Psi_{t+k} \left(\frac{P_t^{*- \epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} \right) \right] \right\}$$

FOC

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[(1-\epsilon) \frac{P_t^{*- \epsilon}}{P_{t+k}^{-\epsilon}} C_{t+k} - \Psi'_{t+k|t} (-\epsilon) \frac{P_t^{*- \epsilon - 1}}{P_{t+k}^{-\epsilon}} C_{t+k} \right] \right\} = 0$$

where $\Psi'_{t+k|t}$ denotes the derivative of Ψ_{t+k} evaluated at $Y_{t+k|t}$.

Then:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} \left[(1-\epsilon) Y_{t+k|t} + \psi_{t+k|t} \epsilon P_t^{*-1} Y_{t+k|t} \right] \right\} = 0$$

where $\psi_{t+k|t} \equiv \Psi'_{t+k|t}$.

Then:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} (1-\epsilon) P_t^{*-1} \left[P_t^* + \psi_{t+k|t} \frac{\epsilon}{1-\epsilon} \right] \right\} = 0$$

Then:

$$(1-\epsilon) P_t^{*-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} \psi_{t+k|t} \right] \right\} = 0$$

Finally:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} \psi_{t+k|t} \right] \right\} = 0$$

From the expression above we get (recall $\epsilon > 1$):

$$P_t^* = \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \right\}}$$

Multiplying and dividing the expression above by $\psi_{t|t}$ we get:

$$P_t^* = \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}} \psi_{t|t}$$

That is:

$$P_t^* = \mathcal{M}_t \psi_{t|t}$$

with

$$\mathcal{M}_t \equiv \frac{\epsilon}{\epsilon - 1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$$

Remark. $\mathcal{M}_t$ is a markup over current marginal cost. Notice that the markup is endogenous, and depends not only on the current values of certain variables but also on the expected value of future variables.

The markup can be rewritten as follows:

$$\mathcal{M}_t = \frac{\epsilon}{\epsilon - 1} \frac{Q_{t,t} Y_{t|t} \psi_{t|t} + \sum_{k=1}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{Q_{t,t} Y_{t|t} \psi_{t|t} + \sum_{k=1}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$$

When prices are fully flexible ($\theta = 0$) we get:

$$\mathcal{M}_t^{flex} = \frac{\epsilon}{\epsilon - 1}$$

Then:

$$P_t^{*,flex} = \frac{\epsilon}{\epsilon - 1} \psi_{t|t}$$

which is the standard expression for the optimal price as a markup over marginal cost when prices are flexible and the elasticity of demand is constant.

Price-Level Dynamics

Let ω_k denote the probability that a firm adjusted its price k periods ago. That is, for any period t , ω_k is the probability that a firm adjusted its price for the last time in $t - k$. Then:

$$\omega_k = (1 - \theta) \theta^k$$

Remark: $\sum_{k=0}^{\infty} \omega_k = \sum_{k=0}^{\infty} (1 - \theta) \theta^k = (1 - \theta) \sum_{k=0}^{\infty} \theta^k = (1 - \theta) \frac{1}{1 - \theta} = 1$.

Since there's a continuum of producers uniformly distributed over $[0, 1]$, ω_k gives the fraction of producers that adjusted prices k periods ago. Also, recall that all firms that adjust their prices in a given period set the same price. Then (ordering the firms appropriately), we have:

$$\begin{aligned} P_t &= \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\int_0^{\omega_0} P_t^{*1-\epsilon} di + \int_{\omega_0}^{\omega_0+\omega_1} P_{t-1}^{*1-\epsilon} di + \int_{\omega_0+\omega_1}^{\omega_0+\omega_1+\omega_2} P_{t-2}^{*1-\epsilon} di + \dots \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\omega_0 P_t^{*1-\epsilon} + \omega_1 P_{t-1}^{*1-\epsilon} + \omega_2 P_{t-2}^{*1-\epsilon} + \dots \right]^{\frac{1}{1-\epsilon}} \\ &= \left[\sum_{k=0}^{\infty} \omega_k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \end{aligned}$$

$$\begin{aligned}
P_t &= \left[\sum_{k=0}^{\infty} (1-\theta) \theta^k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \sum_{k=1}^{\infty} (1-\theta) \theta^k P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta \sum_{k=1}^{\infty} (1-\theta) \theta^{k-1} P_{t-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta \sum_{k=0}^{\infty} (1-\theta) \theta^k P_{t-1-k}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\
&= \left[(1-\theta) P_t^{*1-\epsilon} + \theta P_{t-1}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}
\end{aligned}$$

Then:

$$P_t = \left[\theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$$

Notice that $\theta = 0$ (flex. prices) implies $P_t^{flex} = P_t^{*,flex}$.

For future reference, define:

$$\Pi_t \equiv \frac{P_t}{P_{t-1}} \quad (\text{gross inflation rate})$$

$$\pi_t \equiv \ln \Pi_t = \ln P_t - \ln P_{t-1} \quad (\text{inflation rate})$$

Market Clearing

$$N_t = N_t^d \quad \text{where } N_t^d \equiv \int_0^1 N_t(i) di$$

$$M_t = M_t^d$$

$$B_t = 0$$

Remark: $Y_t(i) = C_t(i) \forall i \in [0, 1]$ has already been imposed in the profit-max problem.

Define:

$$Y_t \equiv \left(\int_0^1 Y_t(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$$

Then, $Y_t(i) = C_t(i) \forall i$ implies

$$Y_t = C_t$$

In addition, since $P_t C_t = \int_0^1 P_t(i) C_t(i) di$ we get:

$$P_t Y_t = \int_0^1 P_t(i) Y_t(i) di$$

Notice that the expression above holds even though $Y_t \neq \int_0^1 Y_t(i) di$.

Aggregate Output

From $Y_t(i) = A_t N_t(i)^{1-\alpha}$ we get: $N_t(i) = \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}}$.

Then:

$$\begin{aligned} N_t &= \int_0^1 N_t(i) di \\ &= \int_0^1 \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}} di \\ &= A_t^{\frac{-1}{1-\alpha}} \int_0^1 Y_t(i)^{\frac{1}{1-\alpha}} di \end{aligned}$$

Using $Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} Y_t$ we get:

$$\begin{aligned} N_t &= A_t^{\frac{-1}{1-\alpha}} \int_0^1 \left[\left(\frac{P_t(i)}{P_t} \right)^{-\epsilon} Y_t \right]^{\frac{1}{1-\alpha}} di \\ &= A_t^{\frac{-1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \end{aligned}$$

Then:

$$Y_t = A_t N_t^{1-\alpha} \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{-(1-\alpha)}$$

Then:

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1}$$

where

$$D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$$

captures price dispersion across goods/firms.

Walras Law

Aggregate profits are:

$$\begin{aligned}\Omega_t &\equiv \int_0^1 \Omega_t(i) di = \int_0^1 [P_t(i)Y_t(i) - W_t N_t(i)] di \\ &= \int_0^1 P_t(i)Y_t(i) di - W_t \int_0^1 N_t(i) di \\ &= P_t Y_t - W_t N_t^d\end{aligned}$$

Substituting the expression above and $T_t = M_t - M_{t-1}$ into the household's budget constraint we get:

$$\begin{aligned}P_t C_t + B_t + M_t^d &= W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + \Omega_t + T_t \\ &= W_t N_t + (1 + i_{t-1}) B_{t-1} + M_{t-1} + P_t Y_t - W_t N_t^d \\ &\quad + M_t - M_{t-1} \\ &= W_t N_t + P_t Y_t - W_t N_t^d + M_t \quad \text{imposing } B_{t-1} = 0\end{aligned}$$

Then:

$$P_t(C_t - Y_t) + B_t + (M_t^d - M_t) + W_t(N_t^d - N_t) = 0$$

Nonstochastic Steady State

We look for a *symmetric* nonstochastic steady state with constant money supply and no inflation.

$$M_t = M_{t+1} = \overline{M}$$

$$P_t = P_{t+1} = \overline{P}$$

Symmetry implies:

$$\overline{P}(i) = \overline{P}(j) \quad \forall i, j \in [0, 1]$$

Substituting into the definition of P_t we get:

$$\overline{P} = \overline{P}(i)$$

Then: $\overline{\Pi} = \frac{\overline{P}}{\overline{P}}$ gives:

$$\overline{\Pi} = 1$$

and $\bar{\pi} = \ln \bar{P} - \ln \bar{P}$ yields:

$$\bar{\pi} = 0$$

Substituting into $C_t(i) = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\epsilon} C_t(j)$ we get:

$$\bar{C}(i) = \bar{C}(j)$$

Substituting into the definition of C_t we get:

$$\bar{C} = \bar{C}(i)$$

Since $Y_t(i) = C_t(i)$ we get:

$$\bar{Y}(i) = \bar{C}(i)$$

$$\bar{Y}(i) = \bar{Y}(j)$$

$$\bar{Y} = \bar{Y}(i)$$

$$\bar{Y} = \bar{C}$$

From $Y_t(i) = A_t N_t(i)^{1-\alpha}$ and $\bar{Y} = \bar{Y}(i)$ we get:

$$\bar{N}(i) = \bar{N}(j)$$

and then $N_t = \int_0^1 N_t(i) di$ gives

$$\bar{N} = \bar{N}(i)$$

From $\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$ we get:

$$\bar{A} = A$$

From $D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$ we get:

$$\bar{D} = 1$$

Then, $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ gives:

$$\bar{Y} = A \bar{N}^{1-\alpha}$$

From $\bar{P} = \left[\theta \bar{P}^{1-\epsilon} + (1-\theta) \bar{P}^{*1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$ we get:

$$\bar{P}^* = \bar{P}$$

From $1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$ we get:

$$1 + \bar{i} = \frac{1}{\beta}$$

For future reference, define:

$$\bar{i}_t \equiv \ln(1 + i_t)$$

Then:

$$\begin{aligned} \bar{i} &= \ln(1 + \bar{i}) \\ &= \ln\left(\frac{1}{\beta}\right) = -\ln \beta \equiv \rho \end{aligned}$$

From $\frac{M_t}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1+i_t} \right)^{-\frac{1}{\nu}}$ we get:

$$\frac{\bar{M}}{\bar{P}} = \chi^{\frac{1}{\nu}} \bar{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}$$

From $C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$ we get:

$$\frac{\overline{W}}{\overline{P}} = \overline{C}^\sigma \overline{N}^\varphi$$

The expression above and $\overline{C} = \overline{Y} = A\overline{N}^{1-\alpha}$ give:

$$\frac{\overline{W}}{\overline{P}} = A^\sigma \overline{N}^{(1-\alpha)\sigma + \varphi}$$

From $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t(i)^{\frac{\alpha}{1-\alpha}}$ and $Y_t(i) = A_t N_t(i)^{1-\alpha}$ we get $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-1} N_t(i)^\alpha$. Then:

$$\overline{\psi} = \frac{1}{1-\alpha} \frac{\overline{W} \overline{N}^\alpha}{A}$$

From $\mathcal{M}_t \equiv \frac{\epsilon}{\epsilon-1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t|t} \right\}}$ we get:

$$\overline{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$$

From $P_t^* = \mathcal{M}_t \psi_{t|t}$, $\bar{P}^* = \bar{P}$, and $\bar{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$ we get:

$$\bar{P} = \frac{\epsilon}{\epsilon-1} \bar{\psi}$$

Using $\bar{\psi} = \frac{1}{1-\alpha} \frac{\bar{W} \bar{N}^\alpha}{A}$ into the previous expression gives:

$$\frac{\bar{W}}{\bar{P}} = \frac{\epsilon-1}{\epsilon} (1-\alpha) A \bar{N}^{-\alpha}$$

Combining the expression above with $\frac{\bar{W}}{\bar{P}} = A^\sigma \bar{N}^{(1-\alpha)\sigma+\varphi}$ we get: $A^\sigma \bar{N}^{(1-\alpha)\sigma+\varphi} = \frac{\epsilon-1}{\epsilon} (1-\alpha) A \bar{N}^{-\alpha}$. Then:

$$\bar{N} = \left[\frac{\epsilon-1}{\epsilon} (1-\alpha) A^{1-\sigma} \right]^{\frac{1}{\alpha+\sigma(1-\alpha)+\varphi}}$$

Having found $\bar{N}$ we can use our previous expressions to find the values of the other variables in the nonstochastic steady state.

Notice that $\overline{N}$ is independent of $\overline{M}$.

Therefore, all real variables are independent of $\overline{M}$ (e.g., $\overline{C} = \overline{Y} = A\overline{N}^{1-\alpha}$).

From $\frac{\overline{M}}{\overline{P}} = \chi^{\frac{1}{\nu}} \overline{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}$ we get:

$$\overline{P} = \frac{1}{\chi^{\frac{1}{\nu}} \overline{C}^{\frac{\sigma}{\nu}} (1 - \beta)^{-\frac{1}{\nu}}} \overline{M}$$

Since $\overline{C}$ is independent of $\overline{M}$, the expression above shows that $\overline{P}$ is proportional to $\overline{M}$.

Hence, money is neutral in the long run.

Log-linearization

For any variable X define the period- t log-deviation from the nonstochastic steady state as follows:

$$\widetilde{X}_t \equiv \ln X_t - \ln \bar{X}$$

Then:

$$\begin{aligned} X_t &= \bar{X} e^{\widetilde{X}_t} \\ &\cong \bar{X} (1 + \widetilde{X}_t) \end{aligned}$$

For $i_t \equiv \ln(1 + i_t)$ define:

$$\widetilde{i}_t \equiv i_t - \bar{i}$$

Notice that: $\widetilde{i}_t \equiv i_t - \bar{i} = \ln(1 + i_t) - \ln(1 + \bar{i}) \cong i_t - \bar{i}$.

Then:

$$\widetilde{i}_t \cong i_t - \bar{i}$$

From $\ln A_{t+1} = (1 - \rho_a) \ln A + \rho_a \ln A_t + \varepsilon_{t+1}^A$ we get:

$$\widetilde{A}_{t+1} = \rho_a \widetilde{A}_t + \varepsilon_{t+1}^A$$

From $\Pi_t \equiv \frac{P_t}{P_{t-1}}$ we get: $\ln \Pi_t = \ln P_t - \ln P_{t-1}$ and $\ln \bar{\Pi} = \ln \bar{P} - \ln \bar{P}$. Then: $\tilde{\Pi}_t \equiv \ln \Pi_t - \ln \bar{\Pi} = (\ln P_t - \ln \bar{P}) - (\ln P_{t-1} - \ln \bar{P})$. Then:

$$\tilde{\Pi}_t = \tilde{P}_t - \tilde{P}_{t-1}$$

Notice that $\tilde{P}_t - \tilde{P}_{t-1} = (\ln P_t - \ln \bar{P}) - (\ln P_{t-1} - \ln \bar{P}) = \ln P_t - \ln P_{t-1} = \ln \left(\frac{P_t}{P_{t-1}} \right) = \ln \Pi_t \equiv \pi_t$. Then:

$$\begin{aligned} \tilde{\Pi}_t &= \ln \Pi_t \\ &= \pi_t \end{aligned}$$

Hence:

$$\pi_t = \tilde{P}_t - \tilde{P}_{t-1}$$

From $C_t = Y_t$ we get:

$$\tilde{C}_t = \tilde{Y}_t$$

Recall $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ with $D_t \equiv \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]^{(1-\alpha)}$.

Then: $\ln Y_t = \ln A_t + (1-\alpha) \ln N_t - \ln D_t$ and $\ln \bar{Y} = \ln \bar{A} + (1-\alpha) \ln \bar{N} - \ln \bar{D}$. Subtracting the second from the first we get:

$$\tilde{Y}_t = \tilde{A}_t + (1-\alpha)\tilde{N}_t - \tilde{D}_t$$

where

$$\begin{aligned} \tilde{D}_t &\equiv \ln D_t - \ln \bar{D} \\ &= \ln D_t \quad \text{since } \bar{D} = 1 \\ &= (1-\alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \\ &\equiv d_t \end{aligned}$$

A first-order Taylor approximation of d_t around the symmetric nss with zero inflation gives (see the Appendix):

$$\tilde{D}_t = d_t \cong 0$$

Then, the expression for aggregate output becomes:

$$\tilde{Y}_t = \tilde{A}_t + (1-\alpha)\tilde{N}_t$$

From $P_t = [\theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^*]^{1-\epsilon}$ and $\bar{P} = \bar{P}^*$ we get:

$$P_t^{1-\epsilon} = \theta P_{t-1}^{1-\epsilon} + (1-\theta) P_t^{*1-\epsilon}$$

$$\left(\bar{P} e^{\tilde{P}_t}\right)^{1-\epsilon} = \theta \left(\bar{P} e^{\tilde{P}_{t-1}}\right)^{1-\epsilon} + (1-\theta) \left(\bar{P}^* e^{\tilde{P}_t^*}\right)^{1-\epsilon}$$

$$\bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_t} = \theta \bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_{t-1}} + (1-\theta) \bar{P}^{1-\epsilon} e^{(1-\epsilon)\tilde{P}_t^*} \quad \text{using } \bar{P}^* = \bar{P}$$

$$e^{(1-\epsilon)\tilde{P}_t} = \theta e^{(1-\epsilon)\tilde{P}_{t-1}} + (1-\theta) e^{(1-\epsilon)\tilde{P}_t^*}$$

$$1 + (1-\epsilon)\tilde{P}_t = \theta [1 + (1-\epsilon)\tilde{P}_{t-1}] + (1-\theta) [1 + (1-\epsilon)\tilde{P}_t^*]$$

$$(1-\epsilon)\tilde{P}_t = \theta (1-\epsilon)\tilde{P}_{t-1} + (1-\theta)(1-\epsilon)\tilde{P}_t^*$$

Then:

$$\tilde{P}_t = \theta \tilde{P}_{t-1} + (1-\theta)\tilde{P}_t^*$$

From $C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}$ we get $\sigma \ln C_t + \varphi \ln N_t = \ln W_t - \ln P_t$ and $\sigma \ln \bar{C} + \varphi \ln \bar{N} = \ln \bar{W} - \ln \bar{P}$. Subtracting the second from the first:

$$\sigma \tilde{C}_t + \varphi \tilde{N}_t = \tilde{W}_t - \tilde{P}_t$$

From $1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} (1 + i_t) \right\}$, $i_t \equiv \ln(1 + i_t)$ and $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$ we get:

$$1 = \mathbb{E}_t \left\{ \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{e^{i_t}}{\Pi_{t+1}} \right\}$$

The term inside curly brackets gives: $\ln \beta - \sigma \ln C_{t+1} + \sigma \ln C_t - \ln \Pi_{t+1} + i_t$, and $\ln \beta - \sigma \ln \bar{C} + \sigma \ln \bar{C} - \ln \bar{\Pi} + \bar{i}$. Subtracting the second from the first: $-\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t - \tilde{\Pi}_{t+1} + \tilde{i}_t$. Then, using $\tilde{\Pi}_{t+1} = \pi_{t+1}$, we get:

$$0 = \mathbb{E}_t \left\{ -\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t + \tilde{i}_t - \pi_{t+1} \right\}$$

From $\frac{M_t}{P_t} = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{i_t}{1+i_t} \right)^{-\frac{1}{\nu}}$, $M_t^r \equiv \frac{M_t}{P_t}$, and $i_t \equiv \ln(1 + i_t)$, we get: $M_t^r = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(\frac{e^{i_t} - 1}{e^{i_t}} \right)^{-\frac{1}{\nu}}$. Then:

$$M_t^r = \chi^{\frac{1}{\nu}} C_t^{\frac{\sigma}{\nu}} \left(1 - e^{-i_t} \right)^{-\frac{1}{\nu}}$$

From the expression above we get: $\ln M_t^r = \frac{1}{\nu} \ln \chi + \frac{\sigma}{\nu} \ln C_t - \frac{1}{\nu} \ln \left(1 - e^{-i_t} \right)$ and $\ln \bar{M}^r = \frac{1}{\nu} \ln \chi + \frac{\sigma}{\nu} \ln \bar{C} - \frac{1}{\nu} \ln \left(1 - e^{-\bar{i}} \right)$. Subtracting the second from the first: $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \frac{1}{\nu} \left[\ln \left(1 - e^{-i_t} \right) - \ln \left(1 - e^{-\bar{i}} \right) \right]$. A first-order Taylor approximation of $\ln \left(1 - e^{-i_t} \right)$ around $i_t = \bar{i}$ gives: $\ln \left(1 - e^{-i_t} \right) \cong \ln \left(1 - e^{-\bar{i}} \right) + \frac{e^{-\bar{i}}}{1 - e^{-\bar{i}}} \left(i_t - \bar{i} \right)$. Then: $\ln \left(1 - e^{-i_t} \right) - \ln \left(1 - e^{-\bar{i}} \right) \cong \frac{e^{\ln \beta}}{1 - e^{\ln \beta}} \widetilde{i}_t = \frac{\beta}{1 - \beta} \widetilde{i}_t$. Then: $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \frac{\beta}{\nu(1 - \beta)} \widetilde{i}_t$. Then:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \eta \widetilde{i}_t \quad \text{with} \quad \eta \equiv \frac{\beta}{\nu(1 - \beta)}$$

$$\text{From } P_t^* = \frac{\epsilon}{\epsilon-1} \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t} \right\}}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ Q_{t,t+k} Y_{t+k|t} \right\}} \text{ and}$$

$$Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} \text{ we get:}$$

$$\begin{aligned} P_t^* \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \right\} \\ = \frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \psi_{t+k|t} \right\} \end{aligned}$$

Log-linearizing the left-hand side of the expression above we get:

$$\begin{aligned} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k P_t^* \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \right\} \\ = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \bar{P}^* e^{\tilde{P}_t^*} \left(\frac{\bar{C}_e \tilde{C}_{t+k}}{\bar{C}_e \tilde{C}_t} \right)^{-\sigma} \frac{\bar{P}_e \tilde{P}_t}{\bar{P}_e \tilde{P}_{t+k}} \bar{Y}_e \tilde{Y}_{t+k|t} \right\} \end{aligned}$$

$$\begin{aligned}
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^*} \left(e^{\tilde{C}_{t+k} - \tilde{C}_t} \right)^{-\sigma} e^{\tilde{P}_t - \tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^*} e^{-\sigma(\tilde{C}_{t+k} - \tilde{C}_t)} e^{\tilde{P}_t - \tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{\tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t}} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\}
\end{aligned}$$

Log-linearizing the right-hand side of the expression above we get:

$$\begin{aligned}
&\frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left\{ \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}} Y_{t+k|t} \psi_{t+k|t} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \left(\frac{\bar{C}_e \tilde{C}_{t+k}}{\bar{C}_e \tilde{C}_t} \right)^{-\sigma} \frac{\bar{P}_e \tilde{P}_t}{\bar{P}_e \tilde{P}_{t+k}} \bar{Y}_e^{\tilde{Y}_{t+k|t}} \bar{\psi}_e^{\tilde{\psi}_{t+k|t}} \right\}
\end{aligned}$$

$$\begin{aligned}
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \left(e^{\tilde{C}_{t+k}-\tilde{C}_t} \right)^{-\sigma} e^{\tilde{P}_t-\tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} e^{\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{-\sigma(\tilde{C}_{t+k}-\tilde{C}_t)} e^{\tilde{P}_t-\tilde{P}_{t+k}} e^{\tilde{Y}_{t+k|t}} e^{\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ e^{-\sigma\tilde{C}_{t+k}+\sigma\tilde{C}_t+\tilde{P}_t-\tilde{P}_{t+k}+\tilde{Y}_{t+k|t}+\tilde{\psi}_{t+k|t}} \right\} \\
&= \frac{\epsilon}{\epsilon-1} \overline{\psi Y} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

Equating both sides, and using $\overline{P} = \frac{\epsilon}{\epsilon-1} \overline{\psi}$:

$$\begin{aligned}
&\overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\} \\
&= \overline{PY} \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma\tilde{C}_{t+k} + \sigma\tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

$$\begin{aligned}
& \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 + \tilde{P}_t^* - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ 1 - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} 1 - \sigma \tilde{C}_{t+k} + \sigma \tilde{C}_t + \tilde{P}_t - \tilde{P}_{t+k} + \tilde{Y}_{t+k|t} + \tilde{\psi}_{t+k|t} \\ -1 - \tilde{P}_t^* + \sigma \tilde{C}_{t+k} - \sigma \tilde{C}_t - \tilde{P}_t + \tilde{P}_{t+k} - \tilde{Y}_{t+k|t} \end{array} \right\} \\
&= \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} - \tilde{P}_t^* \right\} \\
& \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{P}_t^* \right\} = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\} \\
& \tilde{P}_t^* \sum_{k=0}^{\infty} (\theta\beta)^k = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\} \\
& \tilde{P}_t^* \frac{1}{1-\theta\beta} = \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\}
\end{aligned}$$

Then:

$$\tilde{P}_t^* = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \tilde{\psi}_{t+k|t} \right\}$$

Define the individual marginal cost in real terms:

$$MC_{t+k|t} \equiv \frac{\psi_{t+k|t}}{P_{t+k}}$$

Then:

$$\widetilde{MC}_{t+k|t} = \tilde{\psi}_{t+k|t} - \tilde{P}_{t+k}$$

Then, we can write:

$$\tilde{P}_t^* = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} \right\}$$

Subtracting $\tilde{P}_{t-1}$ from both sides:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} \right\} - \tilde{P}_{t-1}$$

Using $\tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t\{\tilde{P}_{t-1}\}$ we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t\left\{\widetilde{MC}_{t+k|t} + \tilde{P}_{t+k} - \tilde{P}_{t-1}\right\}$$

Now we are going to eliminate the individual real marginal cost, $\widetilde{MC}_{t+k|t}$, from the expression above and replace it by the economy's average real marginal cost $\widetilde{MC}_{t+k}$, to be defined below.

From $\psi_{t+k|t} = \frac{1}{1-\alpha} W_{t+k} A_{t+k}^{-\frac{1}{1-\alpha}} Y_{t+k|t}^{\frac{\alpha}{1-\alpha}}$ we get:

$$\tilde{\psi}_{t+k|t} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k|t}$$

Then, $\widetilde{MC}_{t+k|t} = \tilde{\psi}_{t+k|t} - \tilde{P}_{t+k}$ gives:

$$\widetilde{MC}_{t+k|t} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k|t} - \tilde{P}_{t+k}$$

From $Y_t = A_t N_t^{1-\alpha} D_t^{-1}$ we get: $N_t = A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$.

Then: $W_t N_t = W_t A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$. Then, mar-

ginal cost is: $\psi_t = \frac{1}{1-\alpha} W_t A_t^{-\frac{1}{1-\alpha}} Y_t^{\frac{\alpha}{1-\alpha}} D_t^{\frac{1}{1-\alpha}}$. Then:

$\tilde{\psi}_t = \tilde{W}_t - \frac{1}{1-\alpha} \tilde{A}_t + \frac{\alpha}{1-\alpha} \tilde{Y}_t + \frac{1}{1-\alpha} \tilde{D}_t$. Since $\tilde{D}_t \cong 0$, we get:

$$\tilde{\psi}_t = \tilde{W}_t - \frac{1}{1-\alpha} \tilde{A}_t + \frac{\alpha}{1-\alpha} \tilde{Y}_t$$

Defining $MC_t \equiv \frac{\psi_t}{P_t}$ we get, for period t :

$$\widetilde{MC}_t = \tilde{\psi}_t - \tilde{P}_t$$

Combining the previous two expressions we get, for period $t+k$:

$$\widetilde{MC}_{t+k} = \tilde{W}_{t+k} - \frac{1}{1-\alpha} \tilde{A}_{t+k} + \frac{\alpha}{1-\alpha} \tilde{Y}_{t+k} - \tilde{P}_{t+k}$$

Comparing the expressions for $\widetilde{MC}_{t+k|t}$ and $\widetilde{MC}_{t+k}$ we get:

$$\widetilde{MC}_{t+k|t} = \widetilde{MC}_{t+k} + \frac{\alpha}{1-\alpha} (\tilde{Y}_{t+k|t} - \tilde{Y}_{t+k})$$

But $Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\epsilon} Y_{t+k}$ gives:

$$\tilde{Y}_{t+k|t} = -\epsilon \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right) + \tilde{Y}_{t+k}$$

Combining the previous two expressions we get:

$$\widetilde{MC}_{t+k|t} = \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right)$$

Going back to the expression for $\tilde{P}_t^* - \tilde{P}_{t-1}$:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} \right) \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} \\ - \frac{\alpha\epsilon}{1-\alpha} \left(\tilde{P}_t^* - \tilde{P}_{t+k} + \tilde{P}_{t-1} - \tilde{P}_{t-1} \right) \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} \\ -\frac{\alpha\epsilon}{1-\alpha} \left[(\tilde{P}_t^* - \tilde{P}_{t-1}) - (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right] \\ + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \begin{array}{l} \widetilde{MC}_{t+k} - \frac{\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \\ + \frac{\alpha\epsilon}{1-\alpha} (\tilde{P}_{t+k} - \tilde{P}_{t-1}) + \tilde{P}_{t+k} - \tilde{P}_{t-1} \end{array} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \frac{-\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \sum_{k=0}^{\infty} (\theta\beta)^k$$

$$+ (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \left(1 + \frac{\alpha\epsilon}{1-\alpha} \right) (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \frac{-\alpha\epsilon}{1-\alpha} (\tilde{P}_t^* - \tilde{P}_{t-1}) \frac{1}{1 - \theta\beta}$$

$$+ (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1 - \alpha + \alpha\epsilon}{1-\alpha} (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{-\alpha\epsilon}{1-\alpha}(\tilde{P}_t^* - \tilde{P}_{t-1})$$

$$+ (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\left(\tilde{P}_t^* - \tilde{P}_{t-1} \right) \left(1 + \frac{\alpha\epsilon}{1-\alpha} \right) =$$

$$= (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\left(\tilde{P}_t^* - \tilde{P}_{t-1} \right) \frac{1-\alpha+\alpha\epsilon}{1-\alpha} =$$

$$= (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{1-\alpha}{1-\alpha+\alpha\epsilon} (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} + \frac{1-\alpha+\alpha\epsilon}{1-\alpha}(\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1-\theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \frac{1-\alpha}{1-\alpha+\alpha\epsilon} \widetilde{MC}_{t+k} + (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \Theta \widetilde{MC}_{t+k} + (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\}$$

where

$$\Theta \equiv \frac{1 - \alpha}{1 - \alpha + \alpha\epsilon}$$

Then:

$$\begin{aligned} \tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta) \Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + (1 - \theta\beta) \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ (\tilde{P}_{t+k} - \tilde{P}_{t-1}) \right\} \end{aligned}$$

But (see the Appendix):

$$\sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \frac{1}{1 - \theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})$$

Then:

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + (1 - \theta\beta) \mathbb{E}_t \left\{ \frac{1}{1 - \theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

$$\begin{aligned}\tilde{P}_t^* - \tilde{P}_{t-1} &= (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} \\ &\quad + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ (\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) \right\}\end{aligned}$$

Using $\tilde{P}_{t+k} - \tilde{P}_{t+k-1} = \pi_{t+k}$ we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

From the expression above we get:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ (1 - \theta\beta)\Theta \sum_{k=1}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \sum_{k=1}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ (1 - \theta\beta)\Theta \theta\beta \sum_{k=1}^{\infty} (\theta\beta)^{k-1} \mathbb{E}_t \left\{ \widetilde{MC}_{t+k} \right\} + \theta\beta \sum_{k=1}^{\infty} (\theta\beta)^{k-1} \mathbb{E}_t \left\{ \pi_{t+k} \right\}$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ \theta\beta \left[(1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \widetilde{MC}_{t+1+k} \right\} + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_t \left\{ \pi_{t+1+k} \right\} \right]$$

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta)\Theta \widetilde{MC}_t + \pi_t +$$

$$+ \theta\beta \mathbb{E}_t \left\{ (1 - \theta\beta)\Theta \sum_{k=0}^{\infty} (\theta\beta)^k \widetilde{MC}_{t+1+k} + \sum_{k=0}^{\infty} (\theta\beta)^k \pi_{t+1+k} \right\}$$

Using the Law of Iterated Expectations ($\mathbb{E}_t \mathbb{E}_{t+1} X_{t+k} = \mathbb{E}_t X_{t+k}$) we get:

$$\begin{aligned} \tilde{P}_t^* - \tilde{P}_{t-1} = & (1 - \theta\beta) \Theta \widetilde{MC}_t + \pi_t + \\ & + \theta\beta \mathbb{E}_t \left\{ \begin{aligned} & (1 - \theta\beta) \Theta \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_{t+1} \{ \widetilde{MC}_{t+1+k} \} \\ & + \sum_{k=0}^{\infty} (\theta\beta)^k \mathbb{E}_{t+1} \{ \pi_{t+1+k} \} \end{aligned} \right\} \end{aligned}$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = (1 - \theta\beta) \Theta \widetilde{MC}_t + \pi_t + \theta\beta \mathbb{E}_t \{ \tilde{P}_{t+1}^* - \tilde{P}_t \}$$

Now we want to eliminate $\tilde{P}_t^*$ and $\tilde{P}_{t+1}^*$ from the last expression. From $\tilde{P}_t = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^*$ we get:

$$\tilde{P}_t - \tilde{P}_{t-1} = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^* - \tilde{P}_{t-1}$$

$$\tilde{P}_t - \tilde{P}_{t-1} = (1 - \theta) \tilde{P}_t^* - (1 - \theta) \tilde{P}_{t-1}$$

$$\tilde{P}_t - \tilde{P}_{t-1} = (1 - \theta)(\tilde{P}_t^* - \tilde{P}_{t-1})$$

Then:

$$\tilde{P}_t^* - \tilde{P}_{t-1} = \frac{1}{1 - \theta}(\tilde{P}_t - \tilde{P}_{t-1})$$

Substituting into the previous expression we get:

$$\frac{1}{1 - \theta}(\tilde{P}_t - \tilde{P}_{t-1}) = (1 - \theta\beta)\Theta\widetilde{MC}_t + \pi_t + \theta\beta\mathbb{E}_t\left\{\frac{1}{1 - \theta}(\tilde{P}_{t+1} - \tilde{P}_t)\right\}$$

$$(\tilde{P}_t - \tilde{P}_{t-1}) = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + (1 - \theta)\pi_t + \theta\beta\mathbb{E}_t\left\{\tilde{P}_{t+1} - \tilde{P}_t\right\}$$

$$\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + (1 - \theta)\pi_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

$$[1 - (1 - \theta)]\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

Then:

$$\theta\pi_t = (1 - \theta)(1 - \theta\beta)\Theta\widetilde{MC}_t + \theta\beta\mathbb{E}_t\{\pi_{t+1}\}$$

Notice that $\theta = 0 \Rightarrow \widetilde{MC}_t^{flex} = 0$.

Dividing by $\theta > 0$ we get:

$$\pi_t = \lambda \widetilde{MC}_t + \beta \mathbb{E}_t \{ \pi_{t+1} \}$$

where

$$\lambda \equiv \frac{(1-\theta)(1-\theta\beta)\Theta}{\theta}$$

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \beta^n \mathbb{E}_t \pi_{t+n} = 0$, we get:

$$\pi_t = \lambda \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t \{ \widetilde{MC}_{t+k} \}$$

The next step is to show that $\widetilde{MC}_t$ is proportional to a measure of the *output gap* to be defined below.

Remark. This step is not really necessary to solve the model, but it is standard practice to write the model in this way. You should keep in mind, however, that not every model can be written in this form.

From $\sigma\tilde{C}_t + \varphi\tilde{N}_t = \tilde{W}_t - \tilde{P}_t$ and $\tilde{C}_t = \tilde{Y}_t$ we get:
 $\tilde{W}_t - \tilde{P}_t = \sigma\tilde{Y}_t + \varphi\tilde{N}_t$

From $\tilde{Y}_t = \tilde{A}_t + (1 - \alpha)\tilde{N}_t$ we get: $\tilde{N}_t = \frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$

Then: $\tilde{W}_t - \tilde{P}_t = \sigma\tilde{Y}_t + \varphi\frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$

Then:

$$\tilde{W}_t - \tilde{P}_t = \left(\sigma + \frac{\varphi}{1 - \alpha}\right)\tilde{Y}_t - \frac{\varphi}{1 - \alpha}\tilde{A}_t$$

Recall that: $\widetilde{MC}_t = \tilde{W}_t - \tilde{P}_t - \frac{1}{1 - \alpha}\tilde{A}_t + \frac{\alpha}{1 - \alpha}\tilde{Y}_t$.

Then, using the expression above we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi}{1 - \alpha}\right)\tilde{Y}_t - \frac{\varphi}{1 - \alpha}\tilde{A}_t - \frac{1}{1 - \alpha}\tilde{A}_t + \frac{\alpha}{1 - \alpha}\tilde{Y}_t.$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi}{1 - \alpha} + \frac{\alpha}{1 - \alpha}\right)\tilde{Y}_t - \left(\frac{\varphi}{1 - \alpha} + \frac{1}{1 - \alpha}\right)\tilde{A}_t$$

Then:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha}\right)\tilde{Y}_t - \frac{1 + \varphi}{1 - \alpha}\tilde{A}_t$$

Let Y_t^n denote the *natural level of output*, that is, the output level that would prevail under flexible prices ($\theta = 0$). The expression above gives:

$$\widetilde{MC}_t^n = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n - \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

We know from our study of the price-setting problem that, under full flexibility, the optimal price satisfies: $P_t^{*,flex} = \frac{\epsilon}{\epsilon-1} \psi_{t|t}$, with $P_t^{*,flex} = P_t^{flex}$ and $\psi_{t|t}^{flex} = \psi_t^{flex}$.

Then: $P_t^{flex} = \frac{\epsilon}{\epsilon-1} \psi_t^{flex}$. Hence: $\frac{\psi_t^{flex}}{P_t^{flex}} = \frac{\epsilon-1}{\epsilon}$. Since

$MC_t^n \equiv \frac{\psi_t^{flex}}{P_t^{flex}}$, we get:

$$MC_t^n = \frac{\epsilon - 1}{\epsilon}$$

Then:

$$\widetilde{MC}_t^n = 0$$

Then, our previous expression becomes:

$$0 = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n - \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t.$$

Then:

$$\left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n = \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

$$\frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha} \tilde{Y}_t^n = \frac{1 + \varphi}{1 - \alpha} \tilde{A}_t$$

$$\tilde{Y}_t^n = \frac{1 + \varphi}{\sigma(1 - \alpha) + \varphi + \alpha} \tilde{A}_t$$

Then:

$$\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$$

where

$$\psi_y^n = \frac{1 + \varphi}{\sigma(1 - \alpha) + \varphi + \alpha}$$

The expression above gives: $\tilde{A}_t = \frac{\sigma(1-\alpha)+\varphi+\alpha}{1+\varphi} \tilde{Y}_t^n$.

Substituting into our last expression for $\widetilde{MC}_t$ we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \frac{1 + \varphi}{1 - \alpha} \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 + \varphi} \tilde{Y}_t^n$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha} \tilde{Y}_t^n$$

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t - \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{Y}_t^n$$

Then:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) (\tilde{Y}_t - \tilde{Y}_t^n)$$

Define the **output gap** as the log-deviation of output from its natural level:

$$\tilde{y}_t \equiv \ln Y_t - \ln Y_t^n$$

That is, the output gap measures the percent deviation of current output from its hypothetical flexible-price level.

From our analysis of the nss we know $\bar{Y} = A\bar{N}^{1-\alpha}$ with $\bar{N} = \left[\frac{\epsilon-1}{\epsilon} (1-\alpha) A^{1-\sigma} \right]^{\frac{1}{\alpha+\sigma(1-\alpha)+\varphi}}$. Notice that $\bar{Y}$ is independent of θ . Hence, $\bar{Y} = \bar{Y}^n$. This implies:

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n$$

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n + \ln \bar{Y} - \ln \bar{Y}$$

$$\tilde{y}_t = \ln Y_t - \ln Y_t^n + \ln \bar{Y}^n - \ln \bar{Y} \quad \text{using } \bar{Y} = \bar{Y}^n$$

$$\tilde{y}_t = (\ln Y_t - \ln \bar{Y}) - (\ln Y_t^n - \ln \bar{Y}^n)$$

Then:

$$\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$$

Substituting the latter into $\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) (\tilde{Y}_t - \tilde{Y}_t^n)$ we get:

$$\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) \tilde{y}_t$$

Substituting the expression above into $\pi_t = \lambda \widetilde{MC}_t + \beta \mathbb{E}_t \{\pi_{t+1}\}$ we get :

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \{\pi_{t+1}\}$$

where

$$\begin{aligned} \kappa &\equiv \lambda \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)\Theta}{\theta} \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)(1 - \alpha)}{\theta(1 - \alpha + \alpha\epsilon)} \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \\ &= \frac{(1 - \theta)(1 - \theta\beta)}{\theta} \frac{\sigma(1 - \alpha) + \varphi + \alpha}{1 - \alpha + \alpha\epsilon} \end{aligned}$$

The expression above is often referred to as the **New Keynesian Phillips Curve (NKPC)**.

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \beta^n \mathbb{E}_t \pi_{t+n} = 0$, we get:

$$\pi_t = \kappa \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t \{\tilde{y}_{t+k}\}$$

Now we'll derive a dynamic version of the IS curve.

Combining the Euler equation $0 = \mathbb{E}_t \left\{ -\sigma \tilde{C}_{t+1} + \sigma \tilde{C}_t + \tilde{i}_t - \pi_{t+1} \right\}$

with $\tilde{C}_t = \tilde{Y}_t$ we get:

$$0 = \mathbb{E}_t \left\{ -\sigma \tilde{Y}_{t+1} + \sigma \tilde{Y}_t + \tilde{i}_t - \pi_{t+1} \right\}$$

$$\sigma \tilde{Y}_t = \sigma \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \tilde{i}_t + \mathbb{E}_t \left\{ \pi_{t+1} \right\}$$

Then:

$$\tilde{Y}_t = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \left(\tilde{i}_t - \mathbb{E}_t \left\{ \pi_{t+1} \right\} \right)$$

Since $\tilde{i}_t = i_t - \bar{i}$ we can rewrite the expression above as follows:

$$\tilde{Y}_t = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \left[(i_t - \mathbb{E}_t \left\{ \pi_{t+1} \right\}) - \bar{i} \right]$$

Define the *expected real interest rate* as follows:

$$r_t \equiv i_t - \mathbb{E}_t \{ \pi_{t+1} \}$$

Since there's no inflation in our nss, we have:

$$\begin{aligned} \bar{r} &= \bar{i} \\ &= \rho \end{aligned}$$

Then, we can write:

$$\begin{aligned} \tilde{Y}_t &= \mathbb{E}_t \{ \tilde{Y}_{t+1} \} - \frac{1}{\sigma} (r_t - \bar{r}) \\ &= \mathbb{E}_t \{ \tilde{Y}_{t+1} \} - \frac{1}{\sigma} \tilde{r}_t \end{aligned}$$

where

$$\begin{aligned} \tilde{r}_t &\equiv r_t - \bar{r} \\ &= i_t - \mathbb{E}_t \{ \pi_{t+1} \} - \bar{i} \\ &= \tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \} \end{aligned}$$

When prices are flexible, the expression above becomes:

$$\begin{aligned}\tilde{Y}_t^n &= \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} (r_t^n - \bar{r}^n) \\ &= \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} \tilde{r}_t^n \quad \text{where } \tilde{r}_t^n \equiv r_t^n - \bar{r}^n\end{aligned}$$

Notice that:

$$\bar{r}^n = \bar{r} = \bar{i} = \rho$$

The expression above implies:

$$\begin{aligned}\tilde{r}_t - \tilde{r}_t^n &= (r_t - \bar{r}) - (r_t^n - \bar{r}^n) \\ &= (r_t - \bar{r}) - (r_t^n - \bar{r}) \\ &= r_t - r_t^n\end{aligned}$$

Now we can write:

$$\tilde{Y}_t - \tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} \right\} - \frac{1}{\sigma} \tilde{r}_t - \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} + \frac{1}{\sigma} \tilde{r}_t^n$$

$$\tilde{Y}_t - \tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1} - \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

Using $\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

which is the *Dynamic IS curve* (DIS).

Using $\tilde{r}_t - \tilde{r}_t^n = r_t - r_t^n$ we can rewrite the DIS curve as follows:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (r_t - r_t^n)$$

Remark. Iterating forward on the expression above, and imposing $\lim_{n \rightarrow \infty} \mathbb{E}_t \tilde{y}_{t+n} = 0$, we get:

$$\tilde{y}_t = -\frac{1}{\sigma} \sum_{k=0}^{\infty} \mathbb{E}_t \{ r_{t+k} - r_{t+k}^n \}$$

Now we find an expression for $\tilde{r}_t^n$.

From $\tilde{Y}_t^n = \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n \right\} - \frac{1}{\sigma} \tilde{r}_t^n$ we get: $\tilde{r}_t^n = \sigma \mathbb{E}_t \left\{ \tilde{Y}_{t+1}^n - \tilde{Y}_t^n \right\}$.

But $\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$ gives: $\tilde{Y}_{t+1}^n - \tilde{Y}_t^n = \psi_y^n (\tilde{A}_{t+1} - \tilde{A}_t)$.

Then:

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \left\{ \tilde{A}_{t+1} - \tilde{A}_t \right\}$$

Using $\tilde{r}_t^n = r_t^n - \bar{r}^n = r_t^n - \rho$:

$$r_t^n = \rho + \sigma \psi_y^n \mathbb{E}_t \left\{ \tilde{A}_{t+1} - \tilde{A}_t \right\}$$

Remark. $\tilde{A}_{t+1} - \tilde{A}_t = (\ln A_{t+1} - \ln A) - (\ln A_t - \ln A) = \ln A_{t+1} - \ln A_t = a_{t+1} - a_t = \Delta a_{t+1}$ (using Galí's notation).

From $\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{C}_t - \widetilde{\eta}_t^i$ and $\widetilde{C}_t = \widetilde{Y}_t$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{Y}_t - \widetilde{\eta}_t^i$$

Using $\widetilde{y}_t = \widetilde{Y}_t - \widetilde{Y}_t^n$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} (\widetilde{y}_t + \widetilde{Y}_t^n) - \widetilde{\eta}_t^i$$

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{y}_t + \frac{\sigma}{\nu} \widetilde{Y}_t^n - \widetilde{\eta}_t^i$$

Using $\widetilde{Y}_t^n = \psi_y^n \widetilde{A}_t$ we get:

$$\widetilde{M}_t^r = \frac{\sigma}{\nu} \widetilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \widetilde{A}_t - \widetilde{\eta}_t^i$$

which is our LM curve.

Before making the policy rules explicit, we derive equations for the real wage and the markup over marginal cost.

Define:

$$w_t \equiv \frac{W_t}{P_t}$$

Then:

$$\bar{w} = \frac{\bar{W}}{\bar{P}}$$

and

$$\tilde{w}_t = \tilde{W}_t - \tilde{P}_t$$

Using $\tilde{W}_t - \tilde{P}_t = (\sigma + \frac{\varphi}{1-\alpha})\tilde{Y}_t - \frac{\varphi}{1-\alpha}\tilde{A}_t$ we get:

$$\tilde{w}_t = (\sigma + \frac{\varphi}{1-\alpha})\tilde{Y}_t - \frac{\varphi}{1-\alpha}\tilde{A}_t$$

To log-linearize the markup over marginal cost, recall that $\overline{\mathcal{M}} = \frac{\epsilon}{\epsilon-1}$ and $P_t^* = \mathcal{M}_t \psi_{t|t}$. Then: $\frac{P_t^*}{P_t} = \mathcal{M}_t \frac{\psi_{t|t}}{P_t}$.
Then: $\frac{P_t^*}{P_t} = \mathcal{M}_t MC_{t|t}$. Then:

$$\mathcal{M}_t = \frac{P_t^*}{P_t} \frac{1}{MC_{t|t}}$$

Then:

$$\widetilde{\mathcal{M}}_t = \widetilde{P}_t^* - \widetilde{P}_t - \widetilde{MC}_{t|t}$$

Using $\widetilde{MC}_{t|t} = \widetilde{MC}_t - \frac{\alpha\epsilon}{1-\alpha}(\widetilde{P}_t^* - \widetilde{P}_t)$ we get: $\widetilde{\mathcal{M}}_t = \widetilde{P}_t^* - \widetilde{P}_t - \widetilde{MC}_t + \frac{\alpha\epsilon}{1-\alpha}(\widetilde{P}_t^* - \widetilde{P}_t)$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right)(\widetilde{P}_t^* - \widetilde{P}_t) - \widetilde{MC}_t$$

We know $\widetilde{MC}_t = \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) \widetilde{y}_t$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right)(\widetilde{P}_t^* - \widetilde{P}_t) - \left(\sigma + \frac{\varphi+\alpha}{1-\alpha}\right) \widetilde{y}_t$$

From $\tilde{P}_t = \theta \tilde{P}_{t-1} + (1 - \theta) \tilde{P}_t^*$ we get $\tilde{P}_t^* - \tilde{P}_t = \frac{\theta}{1-\theta}(\tilde{P}_t - \tilde{P}_{t-1})$. Using $\pi_t = \tilde{P}_t - \tilde{P}_{t-1}$ we get: $\tilde{P}_t^* - \tilde{P}_t = \frac{\theta}{1-\theta}\pi_t$. Then:

$$\widetilde{\mathcal{M}}_t = \left(1 + \frac{\alpha\epsilon}{1-\alpha}\right) \frac{\theta}{1-\theta} \pi_t - \left(\sigma + \frac{\varphi + \alpha}{1-\alpha}\right) \tilde{y}_t$$

Policy Rules

Money-Growth Rule

$$M_t = \mu_t M_{t-1}$$

with

$$\ln \mu_{t+1} = \rho_\mu \ln \mu_t + \varepsilon_{t+1}^\mu, \quad \varepsilon_{t+1}^\mu \sim N(0, \sigma_{\varepsilon^\mu}^2)$$

Notice that the rule above implies that money growth in the nss is zero:

$$\bar{\mu} = 1$$

Also:

$$\tilde{\mu}_{t+1} = \rho_{\mu} \tilde{\mu}_t + \varepsilon_{t+1}^{\mu}$$

where $\tilde{\mu}_t \equiv \ln \mu_t - \ln \bar{\mu} = \ln \mu_t$.

From $M_t = \mu_t M_{t-1}$ we get: $\frac{M_t}{P_t} = \mu_t \frac{M_{t-1}}{\Pi_t P_{t-1}}$. Then:

$$M_t^r = \frac{\mu_t}{\Pi_t} M_{t-1}^r$$

Then: $\tilde{M}_t^r = \tilde{\mu}_t - \tilde{\Pi}_t + \tilde{M}_{t-1}^r$. Then:

$$\tilde{M}_t^r = \tilde{\mu}_t - \pi_t + \tilde{M}_{t-1}^r$$

Taylor Rule

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

with

$$v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v, \quad \varepsilon_{t+1}^v \sim N(0, \sigma_{\varepsilon^v}^2)$$

Notice that the nss value of v is zero:

$$\bar{v} = 0$$

Using $\tilde{i}_t \equiv i_t - \bar{i} = i_t - \rho$ we can rewrite the Taylor Rule as follows:

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

Remark. Taylor (1993) writes the Taylor Rule, for annual data, as follows:

$$i_t = \pi_t + \frac{1}{2} \tilde{y}_t + \frac{1}{2} (\pi_t - \pi^*) + \rho$$

with $\pi^* = 2\%$ and $\rho = 2\%$.

Then (in percentage points): $i_t = \frac{3}{2}\pi_t + \frac{1}{2}\tilde{y}_t + (\rho - \frac{1}{2}\pi^*) = \frac{3}{2}\pi_t + \frac{1}{2}\tilde{y}_t + (2 - \frac{1}{2}2)$. Then:

$$i_t = 1.5\pi_t + \frac{1}{2}\tilde{y}_t + 1$$

If we set $\phi_\pi = 1.5$, $\phi_y = 0.5$ and $\rho = 2\%$ our rule is similar to Taylor's, except that we have $\pi^* = 0$, and we have added a shock v_t . Another difference is that Taylor measures the output gap as the percentage deviation of output from trend (and not as the deviation from the flexible-price output, which is *unobservable*).

Summarizing

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \{ \pi_{t+1} \}$$

NKPC

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

DIS

$$\tilde{M}_t^r = \frac{\sigma}{\nu} \tilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \tilde{A}_t - \eta \tilde{i}_t$$

LM

$$\tilde{r}_t = \tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \}$$

Fisher Equation

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \tilde{A}_{t+1} - \tilde{A}_t \}$$

Natural Int. Rate

$$\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$$

Natural Output

$$\tilde{y}_t = \tilde{Y}_t - \tilde{Y}_t^n$$

Output Gap

$$\tilde{Y}_t = \tilde{A}_t + (1 - \alpha) \tilde{N}_t$$

Output

$$\tilde{w}_t = (\sigma + \frac{\varphi}{1-\alpha}) \tilde{Y}_t - \frac{\varphi}{1-\alpha} \tilde{A}_t$$

Real Wage

$$\tilde{\mathcal{M}}_t = (1 + \frac{\alpha\epsilon}{1-\alpha}) \frac{\theta}{1-\theta} \pi_t - \left(\sigma + \frac{\varphi+\alpha}{1-\alpha} \right) \tilde{y}_t$$

Markup

$$\tilde{A}_{t+1} = \rho_a \tilde{A}_t + \varepsilon_{t+1}^A$$

TFP Shock

Money-Growth Rule

$$\widetilde{M}_t^r = \widetilde{\mu}_t - \pi_t + \widetilde{M}_{t-1}^r$$

$$\widetilde{\mu}_{t+1} = \rho_\mu \widetilde{\mu}_t + \varepsilon_{t+1}^\mu$$

Taylor Rule

$$\widetilde{i}_t = \phi_\pi \pi_t + \phi_y \widetilde{y}_t + v_t$$

$$v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v$$

Under both policy rules we have a dynamic stochastic system of 13 equations in 13 variables: $\pi_t, \tilde{y}_t, \tilde{i}_t, \tilde{r}_t, \tilde{r}_t^n, \tilde{M}_t^r, \tilde{Y}_t^n, \tilde{Y}_t, \tilde{N}_t, \tilde{w}_t, \tilde{\mathcal{M}}_t, \tilde{A}_t$ plus $\tilde{\mu}_t$ (money-growth rule) or v_t (Taylor rule).

We can solve the system by Uhlig's method. We use "brute force" and declare all endogenous variables to be state variables. Define:

$$\begin{aligned} x_t &\equiv \left(\tilde{M}_t^r, \tilde{y}_t, \pi_t, \tilde{i}_t, \tilde{r}_t, \tilde{r}_t^n, \tilde{Y}_t^n, \tilde{Y}_t, \tilde{N}_t, \tilde{w}_t, \tilde{\mathcal{M}}_t \right)' \\ z_t &\equiv \left(\tilde{A}_t, \tilde{\mu}_t \right)', \quad \varepsilon_{t+1} \equiv \left(\varepsilon_{t+1}^A, \varepsilon_{t+1}^\mu \right)' \quad \text{money-growth rule} \\ z_t &\equiv \left(\tilde{A}_t, v_t \right)', \quad \varepsilon_{t+1} \equiv \left(\varepsilon_{t+1}^A, \varepsilon_{t+1}^v \right)' \quad \text{Taylor rule} \end{aligned}$$

Then, we can write the system in the following form:

$$\begin{aligned} \mathbf{0} &= \mathbb{E}_t \{ Fx_{t+1} + Gx_t + Hx_{t-1} + Lz_{t+1} + Mz_t \} \\ z_{t+1} &= Nz_t + \varepsilon_{t+1} \end{aligned}$$

The solution will be of the form:

$$x_t = Px_{t-1} + Qz_t$$

Money-Growth Rule

$$N_{2 \times 2} = \begin{bmatrix} \rho_a & 0 \\ 0 & \rho_\mu \end{bmatrix}$$

$$F_{11 \times 11} = \begin{bmatrix} 0 & 0 & \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$${}_{11 \times 11} G = \begin{bmatrix} 0 & \kappa & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -\frac{1}{\sigma} & \frac{1}{\sigma} & 0 & 0 & 0 & 0 & 0 \\ 1 & -\frac{\sigma}{\nu} & 0 & \eta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & G_{8,9} & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{10,8} & 0 & 1 & 0 \\ 1 & G_{11,2} & G_{11,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$G_{8,9} = -(1 - \alpha)$$

$$G_{10,8} = -(\sigma + \frac{\varphi}{1 - \alpha})$$

$$G_{11,2} = \sigma + \frac{\varphi + \alpha}{1 - \alpha}$$

$$G_{11,3} = -(1 + \frac{\alpha \epsilon}{1 - \alpha}) \frac{\theta}{1 - \theta}$$

$$\begin{aligned}
{}_{11 \times 11} H &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\
{}_{11 \times 2} L &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\sigma\psi_y^n & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad {}_{11 \times 2} M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{\sigma}{\nu}\psi_y^n & 0 \\ 0 & 0 \\ \sigma\psi_y^n & 0 \\ -\psi_y^n & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ \frac{\phi}{1-\alpha} & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

Taylor Rule

$$N_{2 \times 2} = \begin{bmatrix} \rho_a & 0 \\ 0 & \rho_v \end{bmatrix}$$

$$F_{11 \times 11} = \begin{bmatrix} 0 & 0 & \beta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{11 \times 11} = \begin{bmatrix} 0 & \kappa & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & -\frac{1}{\sigma} & \frac{1}{\sigma} & 0 & 0 & 0 & 0 & 0 \\ 1 & -\frac{\sigma}{\nu} & 0 & \eta & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & G_{8,9} & 0 & 0 \\ 0 & -\phi_y & -\phi_\pi & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{10,8} & 0 & 1 & 0 \\ 1 & G_{11,2} & G_{11,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$G_{8,9} = -(1 - \alpha)$$

$$G_{10,8} = -\left(\sigma + \frac{\varphi}{1 - \alpha}\right)$$

$$G_{11,2} = \sigma + \frac{\varphi + \alpha}{1 - \alpha}$$

$$G_{11,3} = -\left(1 + \frac{\alpha \epsilon}{1 - \alpha}\right) \frac{\theta}{1 - \theta}$$

$${}_{11 \times 11} H = [\mathbf{0}] \quad (\text{all zeros})$$

$${}_{11 \times 2} L = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\sigma\psi_y^n & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad {}_{11 \times 2} M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{\sigma}{\nu}\psi_y^n & 0 \\ 0 & 0 \\ \sigma\psi_y^n & 0 \\ -\psi_y^n & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \\ \frac{\phi}{1-\alpha} & 0 \\ 0 & 0 \end{bmatrix}$$

Calibration (quarterly data)

$$\theta = \frac{2}{3}, \alpha = \frac{1}{3}, \beta = 0.99, \sigma = 1, \nu = 1,$$

$$\eta = 4, \varphi = 1, \phi_\pi = 1.5, \phi_y = \frac{0.5}{4}, \chi = 1$$

$$\epsilon = 6, A = 1, \rho_a = 0.9, \rho_\mu = \rho_v = 0.5$$

These values give:

$$\Theta = 0.25, \lambda = 0.0425, \kappa = 0.1275, \psi_y^n = 1$$

Remark. We follow Galí's calibration and set $\eta = 4$ even though this does not satisfy $\eta = \frac{\beta}{\nu(1-\beta)}$ for our values of β and ν .

Solution under Money-growth Rule

$$P = \begin{bmatrix} 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.2961 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.2084 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.0000 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.7039 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.0559 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1.7598 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.2569 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.2407 & 0.4537 \\ -0.7593 & 1.1204 \\ -0.2407 & 0.5463 \\ 0.0000 & 0.1667 \\ 0.1454 & -0.2408 \\ -0.1000 & 0.0000 \\ 1.0000 & 0.0000 \\ 0.2407 & 1.1204 \\ -1.1389 & 1.6806 \\ -0.8982 & 2.8010 \\ 0.3519 & 1.0088 \end{bmatrix}$$

Then:

$$\widetilde{M}_t^r = 0.7039 \widetilde{M}_{t-1}^r + 0.2407 \widetilde{A}_t + 0.4537 \widetilde{\mu}_t$$

$$\widetilde{y}_t = 0.7039 \widetilde{M}_{t-1}^r - 0.7593 \widetilde{A}_t + 1.1204 \widetilde{\mu}_t$$

$$\pi_t = 0.2961 \widetilde{M}_{t-1}^r - 0.2407 \widetilde{A}_t + 0.5463 \widetilde{\mu}_t$$

$$\widetilde{i}_t = 0.1667 \widetilde{\mu}_t$$

$$\widetilde{r}_t = -0.20846 \widetilde{M}_{t-1}^r + 0.1454 \widetilde{A}_t - 0.2408 \widetilde{\mu}_t$$

$$\widetilde{r}_t^n = -0.1 \widetilde{A}_t$$

$$\widetilde{Y}_t^n = \widetilde{A}_t$$

$$\widetilde{Y}_t = 0.7039 \widetilde{M}_{t-1}^r + 0.2407 \widetilde{A}_t + 1.1204 \widetilde{\mu}_t$$

$$\widetilde{N}_t = 1.0559 \widetilde{M}_{t-1}^r - 1.1389 \widetilde{A}_t + 1.6806 \widetilde{\mu}_t$$

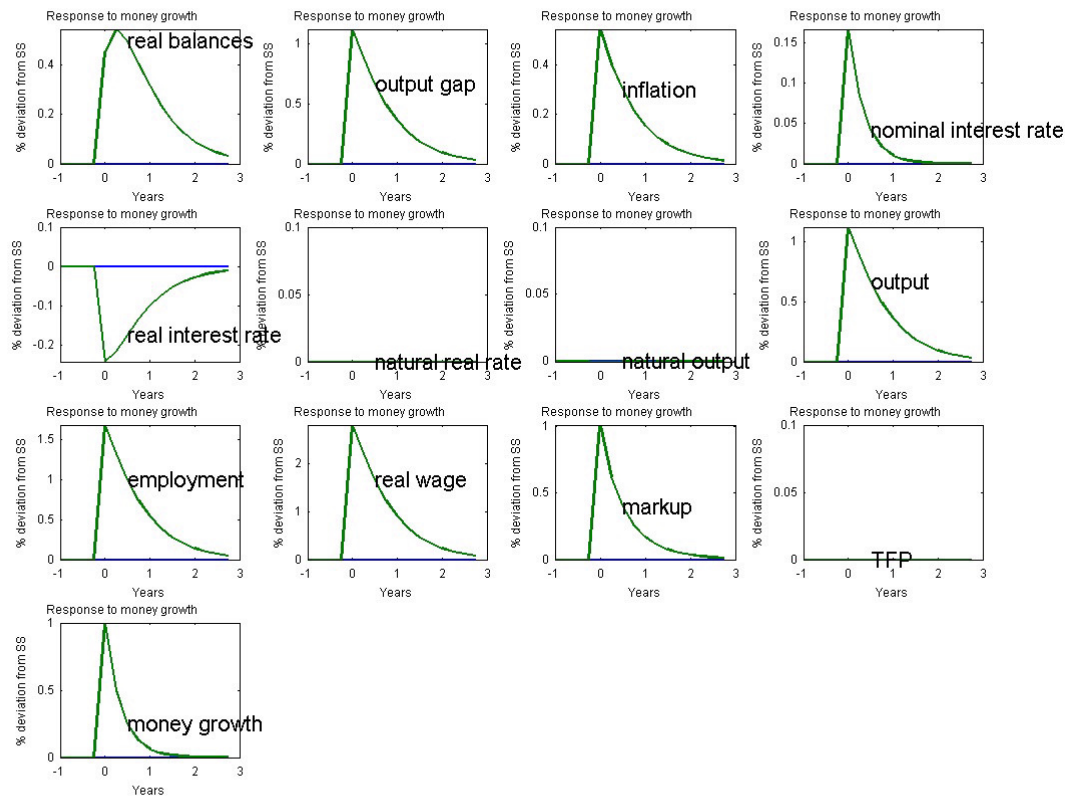
$$\widetilde{w}_t = 1.7598 \widetilde{M}_{t-1}^r - 0.8982 \widetilde{A}_t + 2.8010 \widetilde{\mu}_t$$

$$\widetilde{\mathcal{M}}_t = 0.2569 \widetilde{M}_{t-1}^r + 0.3519 \widetilde{A}_t + 1.0088 \widetilde{\mu}_t$$

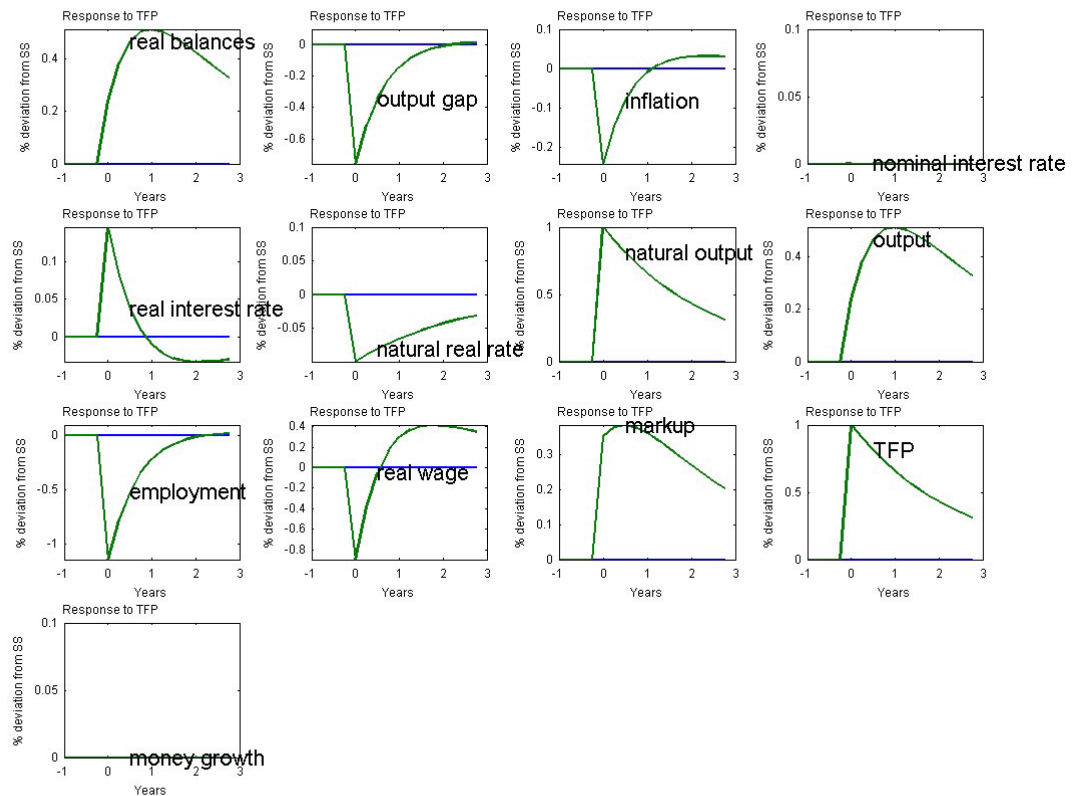
$$\widetilde{A}_{t+1} = 0.90 \widetilde{A}_t + \varepsilon_{t+1}^A$$

$$\widetilde{\mu}_{t+1} = 0.50 \widetilde{\mu}_t + \varepsilon_{t+1}^\mu$$

Impulse Responses to a Money Shock ($\theta = \frac{2}{3}$)



Impulse Responses to a TFP shock ($\theta = \frac{2}{3}$)



Taylor Rule

The solution is:

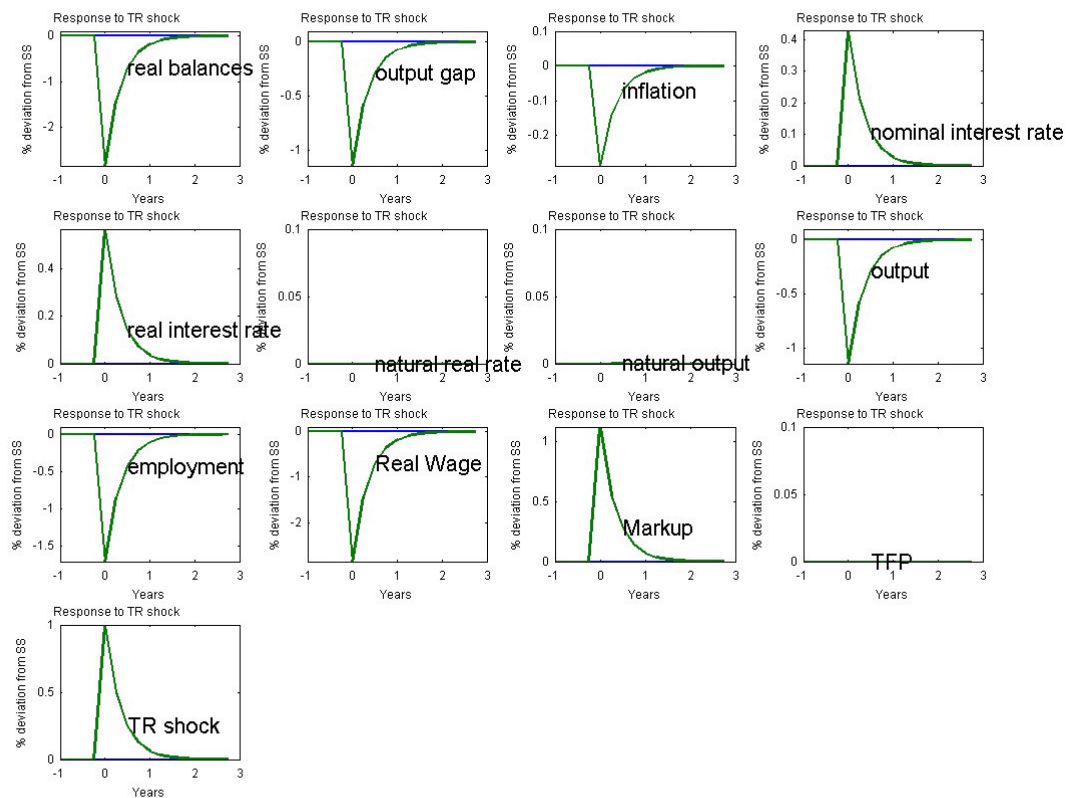
$$P = [0] \quad (\text{all zeros})$$

$$Q = \begin{bmatrix} 1.7033 & -2.8434 \\ -0.1079 & -1.1396 \\ -0.1262 & -0.2877 \\ -0.2028 & 0.4260 \\ -0.0892 & 0.5698 \\ -0.1000 & 0.0000 \\ 1.0000 & 0.0000 \\ 0.8921 & -1.1396 \\ -0.1618 & -1.7094 \\ 0.7303 & -2.8491 \\ -0.6860 & 1.1171 \end{bmatrix}$$

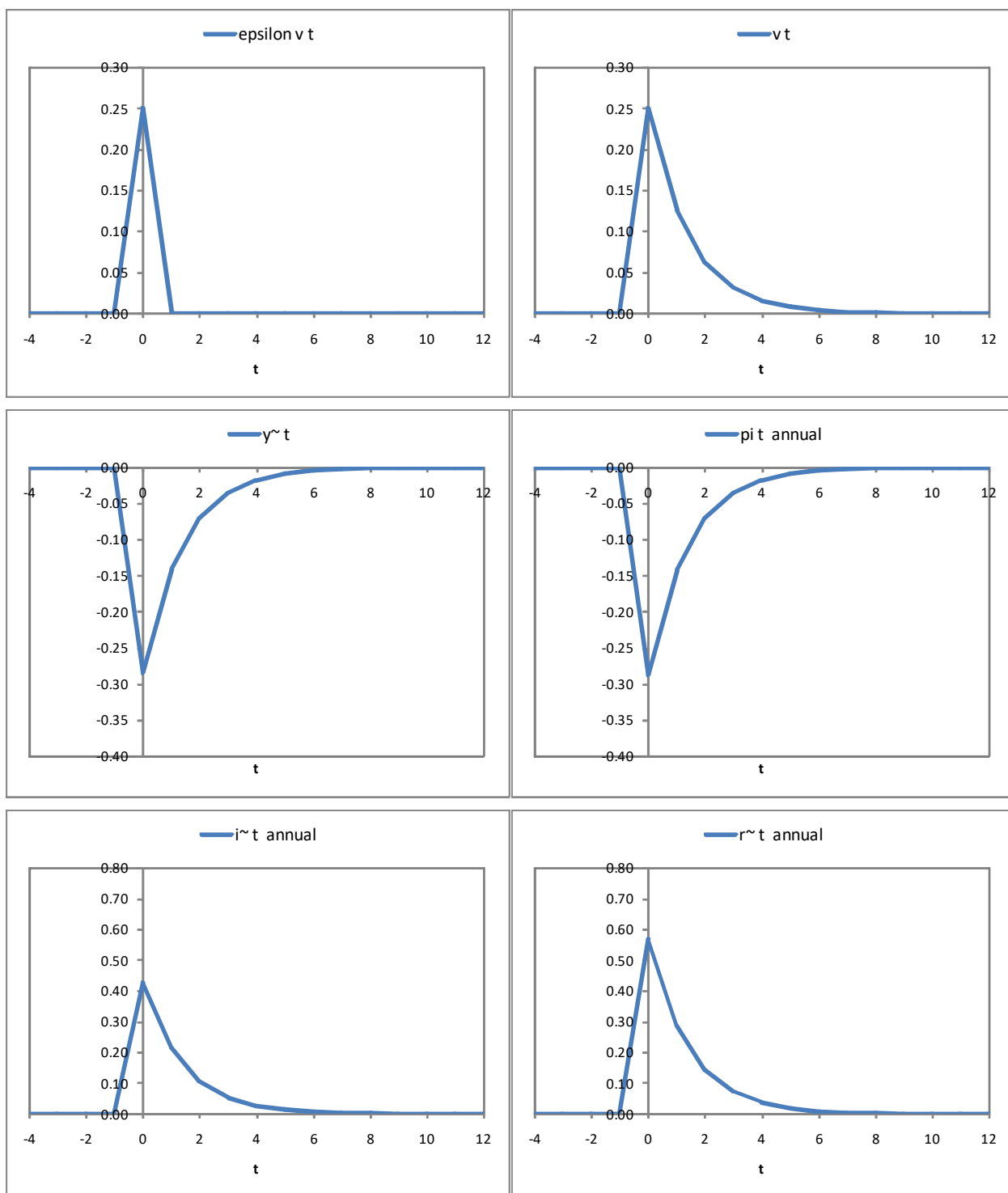
Then:

$$\begin{aligned}\widetilde{M}_t^r &= 1.7033 \widetilde{A}_t - 2.8434 v_t \\ \widetilde{y}_t &= -0.1079 \widetilde{A}_t - 1.1396 v_t \\ \pi_t &= -0.1262 \widetilde{A}_t - 0.2877 v_t \\ \widetilde{i}_t &= -0.2028 \widetilde{A}_t + 0.4260 v_t \\ \widetilde{r}_t &= -0.0892 \widetilde{A}_t + 0.5698 v_t \\ \widetilde{r}_t^n &= -0.1 \widetilde{A}_t \\ \widetilde{Y}_t^n &= \widetilde{A}_t \\ \widetilde{Y}_t &= 0.8921 \widetilde{A}_t - 1.1396 v_t \\ \widetilde{N}_t &= -0.1618 \widetilde{A}_t - 1.7094 v_t \\ \widetilde{w}_t &= 0.7303 \widetilde{A}_t - 2.8491 v_t \\ \widetilde{\mathcal{M}}_t &= -0.6860 \widetilde{A}_t + 1.1171 v_t \\ \widetilde{A}_{t+1} &= 0.90 \widetilde{A}_t + \varepsilon_{t+1}^A \\ v_{t+1} &= 0.50 v_t + \varepsilon_{t+1}^v\end{aligned}$$

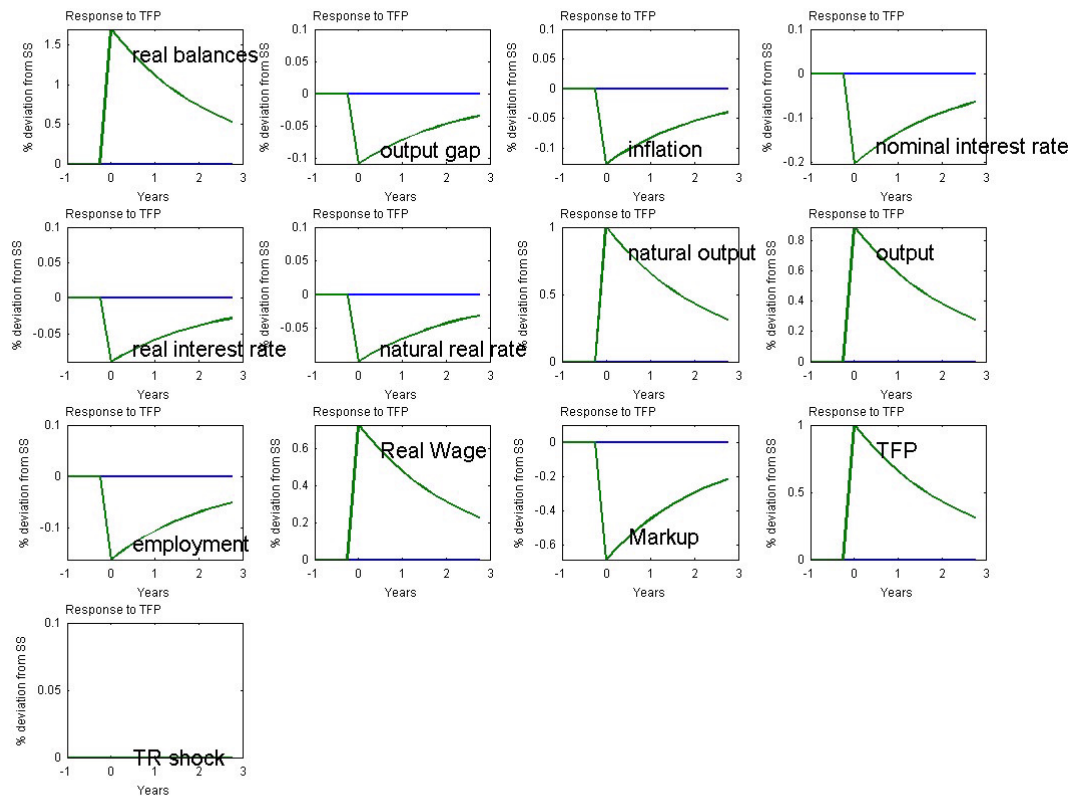
Impulse Responses to a Taylor Rule Shock ($\theta = \frac{2}{3}$)



Galí's Figure 3.1



Impulse Responses to a TFP shock ($\theta = \frac{2}{3}$)



Closed-form Solution and Graphical Representation

We work with the Taylor-Rule version of the model. Analogous results can be obtained using the Money-Growth version.

Strategy: Find a closed-form solution for $\tilde{y}$ and π , and represent the equilibrium with two plots; (i) DIS-MP in $\tilde{y}$ - $\tilde{i}$ space; (ii) DAD-DAS in $\tilde{y}$ - π space

where

DIS: Dynamic IS Curve

MP: Monetary Policy (Taylor Rule) Curve

DAD: Dynamic Aggregate Demand

DAS: Dynamic Aggregate Supply

Closed-Form Solution

From the Taylor Rule we get:

$$\tilde{i}_t = [\phi_\pi \pi_t + v_t] + \phi_y \tilde{y}_t \quad \text{MP}$$

which shows that, for given values of π_t and v_t , there's a positive relationship between $\tilde{i}_t$ and $\tilde{y}_t$.

Combining the original DIS curve and the Fisher Equation we get:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{r}_t - \tilde{r}_t^n)$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} \} - \frac{1}{\sigma} (\tilde{i}_t - \mathbb{E}_t \{ \pi_{t+1} \} - \tilde{r}_t^n)$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} + \frac{1}{\sigma} \tilde{r}_t^n - \frac{1}{\sigma} \tilde{i}_t$$

But:

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \tilde{A}_{t+1} - \tilde{A}_t \}$$

$$\tilde{r}_t^n = \sigma \psi_y^n \mathbb{E}_t \{ \rho_a \tilde{A}_t + \varepsilon_{t+1}^A - \tilde{A}_t \}$$

$$\tilde{r}_t^n = \sigma \psi_y^n [(\rho_a - 1) \tilde{A}_t + \mathbb{E}_t \varepsilon_{t+1}^A]$$

$$\tilde{r}_t^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t$$

Then:

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} + \frac{1}{\sigma} [-\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = \mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} - \psi_y^n (1 - \rho_a) \tilde{A}_t - \frac{1}{\sigma} \tilde{i}_t$$

Then:

$$\tilde{y}_t = [\mathbb{E}_t \{ \tilde{y}_{t+1} + \frac{1}{\sigma} \pi_{t+1} \} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t \quad \text{DIS}$$

which shows that, for given values of $\mathbb{E}_t \tilde{y}_{t+1}$, $\mathbb{E}_t \pi_{t+1}$ and $\tilde{A}_t$, there's a negative relationship between $\tilde{i}_t$ and $\tilde{y}_t$.

Now we can use MP to eliminate $\tilde{i}_t$ from DIS. We get:

$$\tilde{y}_t = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t] - \frac{1}{\sigma}\tilde{i}_t$$

$$\tilde{y}_t = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t] - \frac{1}{\sigma}[\phi_\pi\pi_t + v_t + \phi_y\tilde{y}_t]$$

$$\tilde{y}_t(1 + \frac{\phi_y}{\sigma}) = [\mathbb{E}_t\{\tilde{y}_{t+1} + \frac{1}{\sigma}\pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\phi_\pi}{\sigma}\pi_t$$

$$\tilde{y}_t \frac{\sigma + \phi_y}{\sigma} = [\frac{1}{\sigma}\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\phi_\pi}{\sigma}\pi_t$$

$$\tilde{y}_t = \frac{\sigma}{\sigma + \phi_y} [\frac{1}{\sigma}\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \psi_y^n(1-\rho_a)\tilde{A}_t - \frac{1}{\sigma}v_t] - \frac{\sigma}{\sigma + \phi_y} \frac{\phi_\pi}{\sigma} \pi_t$$

Then:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\mathbb{E}_t\{\sigma\tilde{y}_{t+1} + \pi_{t+1}\} - \sigma\psi_y^n(1-\rho_a)\tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \quad \text{DAD}$$

which shows that, for given values of $\mathbb{E}_t\tilde{y}_{t+1}$, $\mathbb{E}_t\pi_{t+1}$, $\tilde{A}_t$ and v_t , there's a negative relationship between $\tilde{y}_t$ and π_t

From the NKPC we get:

$$\pi_t = \beta \mathbb{E}_t \{\pi_{t+1}\} + \kappa \tilde{y}_t \quad \text{DAS}$$

which shows that, given $\mathbb{E}_t \pi_{t+1}$, there's a positive relationship between $\tilde{y}_t$ and π_t .

Now we can use DAD and DAS to solve for the equilibrium values of π_t and $\tilde{y}_t$. We'll guess the form of the solution and then use the method of undetermined coefficients to obtain the values of the parameters.

Guess:

$$\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$$

$$\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$$

where ψ_{ya} , ψ_{yv} , $\psi_{\pi a}$ and $\psi_{\pi v}$ are coefficients to be determined.

From $\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$ we get:

$$\tilde{y}_{t+1} = \psi_{ya} \tilde{A}_{t+1} + \psi_{yv} v_{t+1}$$

$$\mathbb{E}_t \tilde{y}_{t+1} = \psi_{ya} \mathbb{E}_t \tilde{A}_{t+1} + \psi_{yv} \mathbb{E}_t v_{t+1} \Rightarrow$$

$$\mathbb{E}_t \tilde{y}_{t+1} = \psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t$$

From $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$ we get:

$$\pi_{t+1} = \psi_{\pi a} \tilde{A}_{t+1} + \psi_{\pi v} v_{t+1}$$

$$\mathbb{E}_t \pi_{t+1} = \psi_{\pi a} \mathbb{E}_t \tilde{A}_{t+1} + \psi_{\pi v} \mathbb{E}_t v_{t+1} \Rightarrow$$

$$\mathbb{E}_t \pi_{t+1} = \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t$$

Substituting the expressions for $\tilde{y}_t$ and $\mathbb{E}_t\pi_{t+1}$ into DAS we get:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t$$

$$\pi_t = \beta(\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) + \kappa(\psi_{y a} \tilde{A}_t + \psi_{y v} v_t) \Rightarrow$$

$$\pi_t = (\beta \psi_{\pi a} \rho_a + \kappa \psi_{y a}) \tilde{A}_t + (\beta \psi_{\pi v} \rho_v + \kappa \psi_{y v}) v_t$$

Comparing the expression above with $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$ we get:

$$\psi_{\pi a} = \beta \psi_{\pi a} \rho_a + \kappa \psi_{y a}$$

$$\psi_{\pi v} = \beta \psi_{\pi v} \rho_v + \kappa \psi_{y v}$$

Substituting the expressions for π_t , $\mathbb{E}_t \tilde{y}_{t+1}$, and $\mathbb{E}_t \pi_{t+1}$ into DAD we get:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} \left[\begin{aligned} &\sigma (\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t) + (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) \\ &- \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t \end{aligned} \right]$$

$$- \frac{\phi_\pi}{\sigma + \phi_y} (\psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t)$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a + \psi_{\pi a} \rho_a - \sigma \psi_y^n (1 - \rho_a) - \phi_\pi \psi_{\pi a}}{\sigma + \phi_y} \tilde{A}_t + \frac{\sigma \psi_{yv} \rho_v + \psi_{\pi v} \rho_v - 1 - \phi_\pi \psi_{\pi v}}{\sigma + \phi_y} v_t \Rightarrow$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \tilde{A}_t + \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y} v_t$$

Comparing the expression above with $\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$ we get:

$$\psi_{ya} = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y}$$

$$\psi_{yv} = \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y}$$

Hence, we need to solve the following system of four equations in the four unknowns $\psi_{\pi a}$, $\psi_{\pi v}$, ψ_{ya} , ψ_{yv} :

$$\psi_{\pi a} = \beta \psi_{\pi a} \rho_a + \kappa \psi_{ya} \quad (1)$$

$$\psi_{\pi v} = \beta \psi_{\pi v} \rho_v + \kappa \psi_{yv} \quad (2)$$

$$\psi_{ya} = \frac{\sigma \psi_{ya} \rho_a + (\rho_a - \phi_\pi) \psi_{\pi a} - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \quad (3)$$

$$\psi_{yv} = \frac{\sigma \psi_{yv} \rho_v + (\rho_v - \phi_\pi) \psi_{\pi v} - 1}{\sigma + \phi_y} \quad (4)$$

From (1) we get: $\psi_{\pi a}(1 - \beta\rho_A) = \kappa\psi_{ya}$. Then:

$$\psi_{\pi a} = \frac{\kappa}{1-\beta\rho_A}\psi_{ya} \quad (5)$$

From (2) we get: $\psi_{\pi v}(1 - \beta\rho_v) = \kappa\psi_{yv}$. Then:

$$\psi_{\pi v} = \frac{\kappa}{1-\beta\rho_v}\psi_{yv} \quad (6)$$

From (3) we get:

$$(\sigma + \phi_y)\psi_{ya} = \sigma\rho_a\psi_{ya} - \sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\psi_{\pi a}$$

$$(\sigma + \phi_y - \sigma\rho_a)\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\psi_{\pi a}$$

Using (5):

$$(\sigma + \phi_y - \sigma\rho_a)\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a) + (\rho_a - \phi_\pi)\frac{\kappa}{1-\beta\rho_a}\psi_{ya}$$

$$(\sigma + \phi_y - \sigma\rho_a - \frac{\kappa(\rho_a - \phi_\pi)}{1-\beta\rho_a})\psi_{ya} = -\sigma\psi_y^n(1 - \rho_a)$$

$$\frac{(1-\beta\rho_a)(\sigma+\phi_y-\sigma\rho_a)-\kappa(\rho_a-\phi_\pi)}{1-\beta\rho_a}\psi_{ya}=-\sigma\psi_y^n(1-\rho_a)$$

$$\frac{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)}{1-\beta\rho_a}\psi_{ya}=-\sigma\psi_y^n(1-\rho_a)\Rightarrow$$

$$\psi_{ya} = \frac{-\sigma\psi_y^n(1-\rho_a)(1-\beta\rho_a)}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \quad (7)$$

From (4) we get:

$$(\sigma+\phi_y)\psi_{yv}=\sigma\psi_{yv}\rho_v+(\rho_v-\phi_\pi)\psi_{\pi v}-1$$

$$(\sigma+\phi_y-\sigma\rho_v)\psi_{yv}=(\rho_v-\phi_\pi)\psi_{\pi v}-1$$

Using (6):

$$(\sigma+\phi_y-\sigma\rho_v)\psi_{yv}=(\rho_v-\phi_\pi)\frac{\kappa}{1-\beta\rho_v}\psi_{yv}-1$$

$$(\sigma+\phi_y-\sigma\rho_v-\frac{\kappa(\rho_v-\phi_\pi)}{1-\beta\rho_v})\psi_{yv}=-1$$

$$\frac{(1-\beta\rho_v)(\sigma+\phi_y-\sigma\rho_v)-\kappa(\rho_v-\phi_\pi)}{1-\beta\rho_v}\psi_{yv}=-1$$

$$\frac{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)}{1-\beta\rho_v}\psi_{yv}=-1\Rightarrow$$

$$\psi_{yv} = \frac{-(1-\beta\rho_v)}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \quad (8)$$

Substituting (7) back into (5) we get:

$$\psi_{\pi a} = \frac{\kappa}{1-\beta\rho_a} \frac{-\sigma\psi_y^n(1-\rho_a)(1-\beta\rho_a)}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \Rightarrow$$

$$\psi_{\pi a} = \frac{-\sigma\psi_y^n(1-\rho_a)\kappa}{(1-\beta\rho_a)[\sigma(1-\rho_a)+\phi_y]+\kappa(\phi_\pi-\rho_a)} \quad (9)$$

Finally, substituting (8) back into (6) we get:

$$\psi_{\pi v} = \frac{\kappa}{1-\beta\rho_v} \frac{-(1-\beta\rho_v)}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \Rightarrow$$

$$\psi_{\pi v} = \frac{-\kappa}{(1-\beta\rho_v)[\sigma(1-\rho_v)+\phi_y]+\kappa(\phi_\pi-\rho_v)} \quad (10)$$

Summarizing, the solution is:

$$\tilde{y}_t = \psi_{ya} \tilde{A}_t + \psi_{yv} v_t$$

$$\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$$

with

$$\psi_{ya} = -\sigma \psi_y^n (1 - \rho_a)(1 - \beta \rho_a) \Lambda_a$$

$$\psi_{\pi a} = -\sigma \psi_y^n (1 - \rho_a) \kappa \Lambda_a$$

$$\psi_{yv} = -(1 - \beta \rho_v) \Lambda_v$$

$$\psi_{\pi v} = -\kappa \Lambda_v$$

where:

$$\Lambda_a \equiv \frac{1}{(1 - \beta \rho_a)[\sigma(1 - \rho_a) + \phi_y] + \kappa(\phi_\pi - \rho_a)}$$

$$\Lambda_v \equiv \frac{1}{(1 - \beta \rho_v)[\sigma(1 - \rho_v) + \phi_y] + \kappa(\phi_\pi - \rho_v)}$$

Our calibration gives:

$$\psi_{ya} = -0.1079, \psi_{\pi a} = -0.1262, \psi_{yv} = -1.1396, \psi_{\pi v} = -0.2877$$

Once we've found the solution for π_t and $\tilde{y}_t$ we can use the Taylor Rule to obtain the equilibrium value of the nominal interest rate:

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t$$

Then we can get the real interest rate:

$$\tilde{r}_t = \tilde{i}_t - \mathbb{E}_t \{\pi_{t+1}\}$$

Aggregate output is (recall $\tilde{Y}_t^n = \psi_y^n \tilde{A}_t$):

$$\tilde{Y}_t = \tilde{Y}_t^n + \tilde{y}_t$$

From the production function, $\tilde{Y}_t = \tilde{A}_t + (1 - \alpha)\tilde{N}_t$, we get:

$$\tilde{N}_t = \frac{\tilde{Y}_t - \tilde{A}_t}{1 - \alpha}$$

Finally, we can use the LM equation to get real money balances:

$$M_t^r = \frac{\sigma}{\nu} \tilde{y}_t + \frac{\sigma}{\nu} \psi_y^n \tilde{A}_t - \eta \tilde{i}_t$$

Plots

We know:

$$\tilde{i}_t = (\phi_\pi \pi_t + v_t) + \phi_y \tilde{y}_t \quad \text{MP}$$

$$\tilde{y}_t = [\mathbb{E}_t \tilde{y}_{t+1} + \frac{1}{\sigma} \mathbb{E}_t \pi_{t+1} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t \quad \text{DIS}$$

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \quad \text{DAD}$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad \text{DAS}$$

From our solution we know:

$$\mathbb{E}_t \tilde{y}_{t+1} = \psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t$$

$$\mathbb{E}_t \pi_{t+1} = \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t$$

Then, we can eliminate the expected values of future variables from DIS, DAD and DAS.

From DIS we get:

$$\tilde{y}_t = [\mathbb{E}_t \tilde{y}_{t+1} + \frac{1}{\sigma} \mathbb{E}_t \pi_{t+1} - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t + \frac{1}{\sigma} (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t + \frac{1}{\sigma} \psi_{\pi a} \rho_a \tilde{A}_t + \frac{1}{\sigma} \psi_{\pi v} \rho_v v_t - \psi_y^n (1 - \rho_a) \tilde{A}_t] - \frac{1}{\sigma} \tilde{i}_t$$

$$\tilde{y}_t = [(\psi_{ya} + \frac{1}{\sigma} \psi_{\pi a}) \rho_a - \psi_y^n (1 - \rho_a)] \tilde{A}_t + (\psi_{yv} + \frac{1}{\sigma} \psi_{\pi v}) \rho_v v_t - \frac{1}{\sigma} \tilde{i}_t$$

From DAD we get:

$$\tilde{y}_t = \frac{1}{\sigma + \phi_y} [\sigma \mathbb{E}_t \tilde{y}_{t+1} + \mathbb{E}_t \pi_{t+1} - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t] - \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{\sigma (\psi_{ya} \rho_a \tilde{A}_t + \psi_{yv} \rho_v v_t) + (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t}{\sigma + \phi_y}$$

$$- \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{\sigma \psi_{ya} \rho_a \tilde{A}_t + \sigma \psi_{yv} \rho_v v_t + \psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t - \sigma \psi_y^n (1 - \rho_a) \tilde{A}_t - v_t}{\sigma + \phi_y}$$

$$- \frac{\phi \pi}{\sigma + \phi_y} \pi_t$$

$$\tilde{y}_t = \frac{[\sigma\psi_{ya}\rho_a + \psi_{\pi a}\rho_a - \sigma\psi_y^n(1-\rho_a)]\tilde{A}_t + (\sigma\psi_{yv}\rho_v + \psi_{\pi v}\rho_v - 1)v_t}{\sigma + \phi_y} - \frac{\phi_\pi}{\sigma + \phi_y}\pi_t$$

$$\tilde{y}_t = \frac{(\sigma\psi_{ya} + \psi_{\pi a})\rho_a - \sigma\psi_y^n(1-\rho_a)}{\sigma + \phi_y}\tilde{A}_t + \frac{(\sigma\psi_{yv} + \psi_{\pi v})\rho_v - 1}{\sigma + \phi_y}v_t - \frac{\phi_\pi}{\sigma + \phi_y}\pi_t$$

From DAS we get:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t$$

$$\pi_t = \beta(\psi_{\pi a}\rho_a\tilde{A}_t + \psi_{\pi v}\rho_v v_t) + \kappa \tilde{y}_t$$

Hence, we've arrived to a system that we can plot:

$$\tilde{i}_t = (\phi_\pi \pi_t + v_t) + \phi_y \tilde{y}_t \quad \text{MP}$$

$$\begin{aligned} \tilde{y}_t = & [(\psi_{ya} + \frac{1}{\sigma} \psi_{\pi a}) \rho_a - \psi_y^n (1 - \rho_a)] \tilde{A}_t \\ & + (\psi_{yv} + \frac{1}{\sigma} \psi_{\pi v}) \rho_v v_t - \frac{1}{\sigma} \tilde{i}_t \end{aligned} \quad \text{DIS}$$

$$\begin{aligned} \tilde{y}_t = & \frac{(\sigma \psi_{ya} + \psi_{\pi a}) \rho_a - \sigma \psi_y^n (1 - \rho_a)}{\sigma + \phi_y} \tilde{A}_t \\ & + \frac{(\sigma \psi_{yv} + \psi_{\pi v}) \rho_v - 1}{\sigma + \phi_y} v_t - \frac{\phi_\pi}{\sigma + \phi_y} \pi_t \end{aligned} \quad \text{DAD}$$

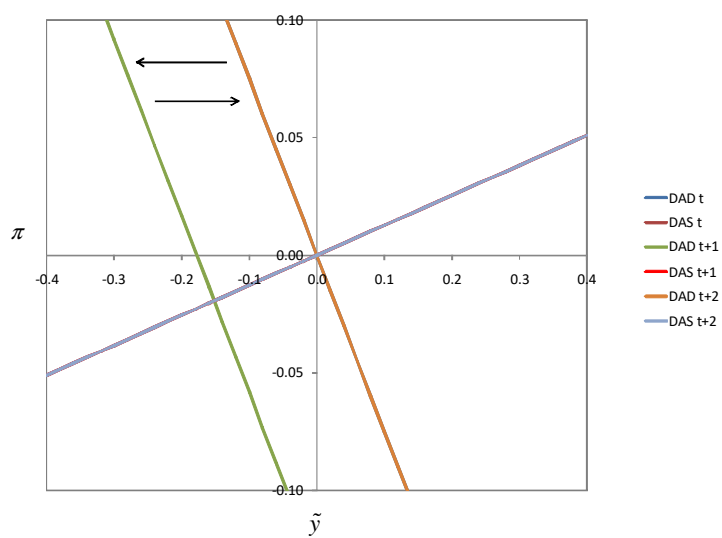
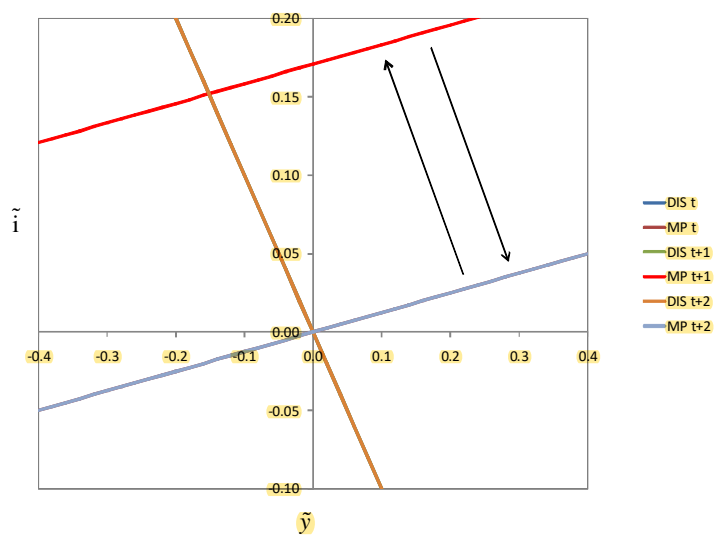
$$\pi_t = \beta (\psi_{\pi a} \rho_a \tilde{A}_t + \psi_{\pi v} \rho_v v_t) + \kappa \tilde{y}_t \quad \text{DAS}$$

The DAD-DAS plot determines $\tilde{y}_t$ and π_t

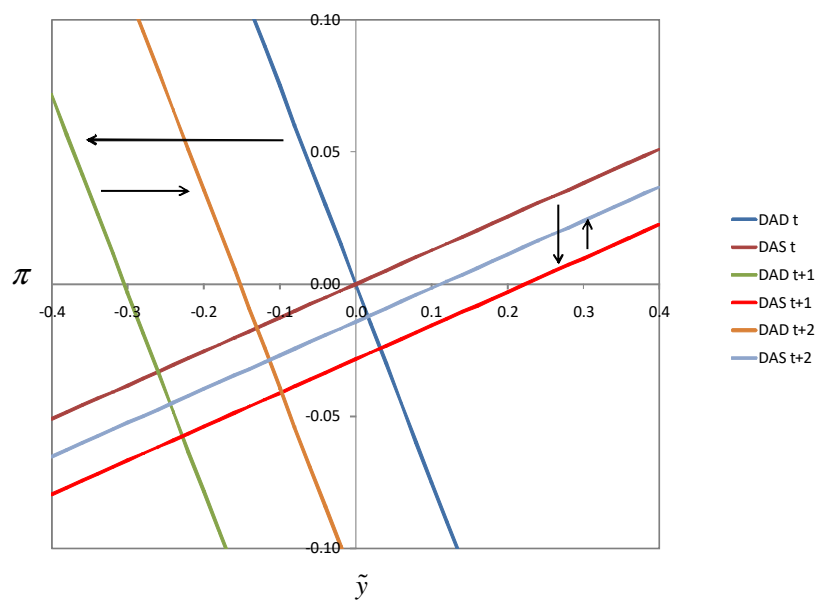
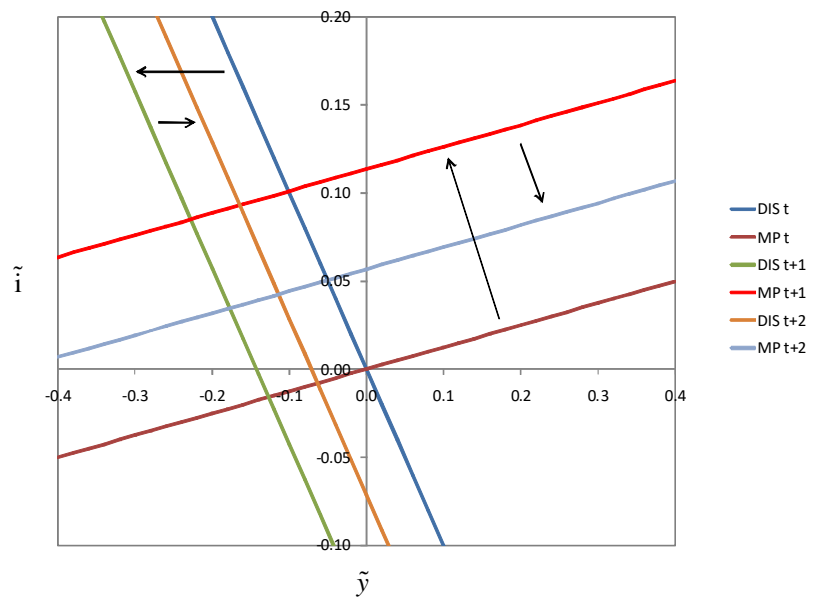
Then π_t determines the position of MP (we can actually eliminate π_t from MP using $\pi_t = \psi_{\pi a} \tilde{A}_t + \psi_{\pi v} v_t$)

Then the DIS-MP plot determines $\tilde{i}_t$

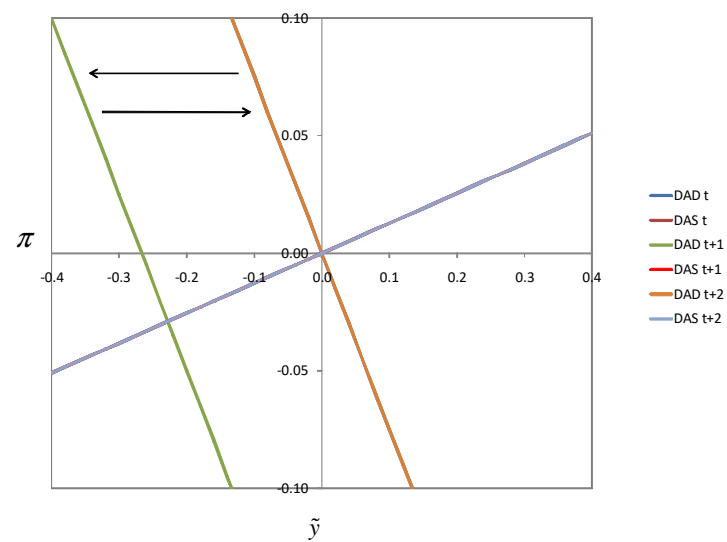
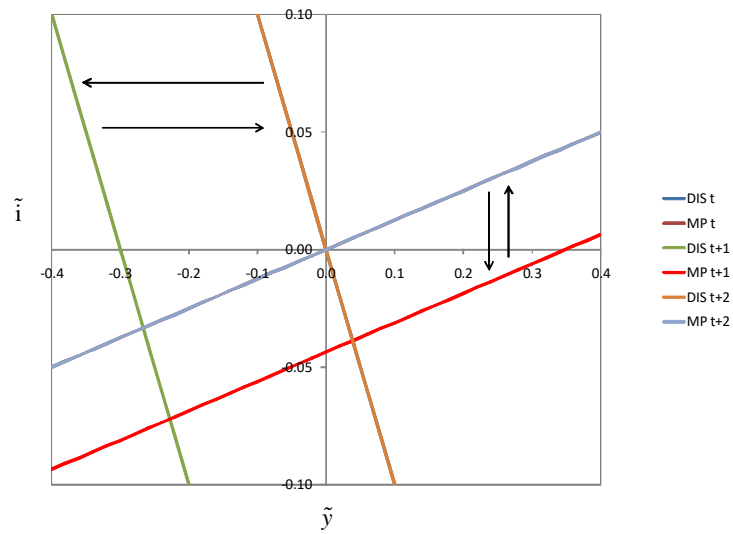
A temporary shock to the Taylor Rule ($\varepsilon_t^v > 0$, $\rho_v = 0$)



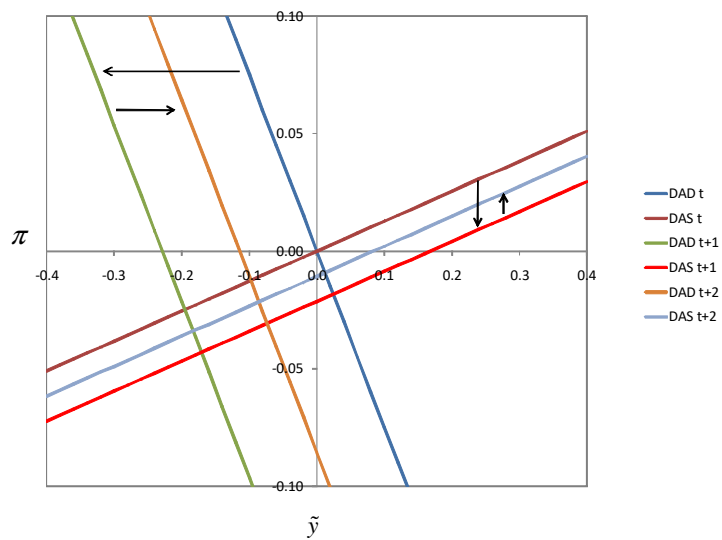
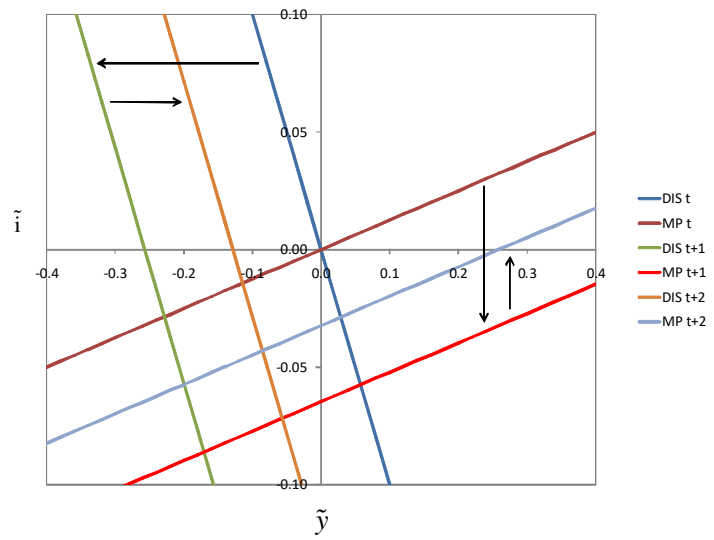
A persistent shock to the Taylor Rule ($\varepsilon_t^v > 0$, $\rho_v > 0$)



A temporary shock to technology ($\varepsilon_t^A > 0, \rho_a = 0$)



A persistent shock to technology ($\varepsilon_t^A > 0, \rho_a > 0$)



Appendix

First-order conditions for the intratemporal problem

Consider the following problem:

$$\begin{aligned} \max \quad & \left(\int_0^1 C(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \\ \text{s.t.} \quad & \int_0^1 P(i)C(i)di = I \end{aligned}$$

Form the Lagrangian:

$$\mathcal{L} = \left(\int_0^1 C(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P(i)C(i)di \right]$$

To derive the first-order conditions we cannot take the pointwise derivative with respect to $C(i)$ for any fixed i , because such derivative is simply zero. Instead, we'll use a variational argument. First notice that what we are choosing is a real function $C(\bullet)$ over the domain $[0, 1]$.

Let $C^*(\bullet)$ be the solution of our problem, and define a new function $C(\bullet, \alpha)$ by perturbing $C^*(\bullet)$, as follows:

$$C(i, \alpha) = C^*(i) + \alpha F(i)$$

where $\alpha \in \mathbb{R}$ and $F(\bullet)$ is an arbitrary (integrable) real function over $[0, 1]$. Notice that, for a given function $F(\bullet)$, each value of α defines a new function. Also, when $\alpha = 0$ we have $C(i, 0) = C^*(i)$.

Now we can write the Lagrangian as a function of α , as follows:

$$\mathcal{L}(\alpha) = \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} + \lambda \left[I - \int_0^1 P(i) C(i, \alpha) di \right]$$

Since our objective is maximized at $\alpha = 0$, which is obviously an interior point of $\mathbb{R}$, the FOC of our problem is $\mathcal{L}'(0) = 0$. From the expression above, and using Leibniz's rule for the differentiation of integrals, we get:

$$\begin{aligned} \mathcal{L}'(\alpha) = & \frac{\epsilon}{\epsilon-1} \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}-1} \int_0^1 \left[\frac{\epsilon-1}{\epsilon} C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}-1} \frac{\partial C(i, \alpha)}{\partial \alpha} \right] di \\ & - \lambda \int_0^1 P(i) \frac{\partial C(i, \alpha)}{\partial \alpha} di \end{aligned}$$

Using $\frac{\partial C(i, \alpha)}{\partial \alpha} = F(i)$ we get:

$$\mathcal{L}'(\alpha) = \left(\int_0^1 C(i, \alpha)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} \int_0^1 C(i, \alpha)^{\frac{-1}{\epsilon}} F(i) di - \lambda \int_0^1 P(i) F(i) di$$

Evaluating at $\alpha = 0$ we get (recall $C(i, 0) = C^*(i)$):

$$\mathcal{L}'(0) = \left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} \int_0^1 C^*(i)^{\frac{-1}{\epsilon}} F(i) di - \lambda \int_0^1 P(i) F(i) di$$

We can rewrite the expression above as follows:

$$\begin{aligned} \mathcal{L}'(0) &= \int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} F(i) \right] di - \int_0^1 \lambda P(i) F(i) di \\ &= \int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) \right] F(i) di \end{aligned}$$

We want to equate the expression above to zero:

$$\int_0^1 \left[\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) \right] F(i) di = 0$$

Since $F(\bullet)$ can be chosen arbitrarily, the only way to satisfy the expression above for any such $F(\bullet)$ is to set the term in brackets equal to zero for each $i \in [0, 1]$. To see this, notice that if the term in brackets is different from zero for some $i \in [0, 1]$, say > 0 , then it will be > 0 over some interval (because of continuity). Hence, choosing an $F(\bullet)$ that is positive over such interval, the product $[\bullet] F(\bullet)$ will be positive over that interval, and then the whole integral will be different from zero. Therefore, we conclude:

$$\left(\int_0^1 C^*(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{1}{\epsilon-1}} C^*(i)^{\frac{-1}{\epsilon}} - \lambda P(i) = 0 \quad \forall i \in [0, 1]$$

Price dispersion

Claim 1. Up to first order, $\int_0^1 [\ln P_t(i) - \ln P_t] di \cong 0$.

Proof.

From $P_t = \left[\int_0^1 P_t(i)^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}}$ we get:

$$P_t^{1-\epsilon} = \int_0^1 P_t(i)^{1-\epsilon} di$$

Dividing by $P_t^{1-\epsilon}$ we get:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} di = 1$$

Notice that $\left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} = e^{\ln \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon}} = e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)}$.

A first-order Taylor approximation of $e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)}$ around $\ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) = \ln(1) = 0$ gives:

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^{(1-\epsilon) \ln(1)} + e^{(1-\epsilon) \ln(1)} (1-\epsilon) \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) \right]$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^0 + e^0 (1-\epsilon) \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln(1) \right]$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 + (1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)$$

$$e^{(1-\epsilon) \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 + (1-\epsilon) [\ln P_t(i) - \ln P_t]$$

Then:

$$\left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} \cong 1 + (1-\epsilon) [\ln P_t(i) - \ln P_t]$$

Then:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\epsilon} di \cong \int_0^1 [1 + (1-\epsilon) (\ln P_t(i) - \ln P_t)] di$$

$$1 \cong \int_0^1 [1 + (1-\epsilon) (\ln P_t(i) - \ln P_t)] di$$

$$1 \cong 1 + (1 - \epsilon) \int_0^1 [\ln P_t(i) - \ln P_t] di$$

$$0 \cong (1 - \epsilon) \int_0^1 [\ln P_t(i) - \ln P_t] di$$

Then:

$$\int_0^1 [\ln P_t(i) - \ln P_t] di \cong 0$$



Claim 2. Up to first order we have $d_t \cong 0$, where $d_t \equiv (1 - \alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right]$.

Proof. Notice that $\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} = e^{\ln \left(\frac{P_t(i)}{P_t} \right) \frac{-\epsilon}{1-\alpha}}$. Then $\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} = e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)}$.

A first-order Taylor approximation of $e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)}$ around $\ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) = \ln(1) = 0$ gives:

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^{-\frac{\epsilon}{1-\alpha} \ln(1)} + e^{-\frac{\epsilon}{1-\alpha} \ln(1)} \frac{-\epsilon}{1-\alpha} \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln \left(\frac{\bar{P}(i)}{\bar{P}} \right) \right]$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong e^0 + e^0 \frac{-\epsilon}{1-\alpha} \left[\ln \left(\frac{P_t(i)}{P_t} \right) - \ln(1) \right]$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 - \frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)$$

$$e^{-\frac{\epsilon}{1-\alpha} \ln \left(\frac{P_t(i)}{P_t} \right)} \cong 1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t]$$

Then:

$$\left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} \cong 1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t]$$

Then:

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong \int_0^1 \left[1 - \frac{\epsilon}{1-\alpha} [\ln P_t(i) - \ln P_t] \right] di$$

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong 1 - \frac{\epsilon}{1-\alpha} \int_0^1 [\ln P_t(i) - \ln P_t] di$$

$$\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \cong 1 \quad \text{by Claim 1.}$$

Then:

$$\ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong \ln(1)$$

Then:

$$\ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong 0$$

Then:

$$d_t \equiv (1-\alpha) \ln \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{\frac{-\epsilon}{1-\alpha}} di \right] \cong (1-\alpha) 0$$

Then:

$$d_t \cong 0.$$



Intermediate step in the derivation of the NKPC

Claim 3. The following equality holds:

$$\sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \frac{1}{1-\theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})$$

Proof. We have:

$$\begin{aligned} & \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t-1}) = \\ &= \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1} + \tilde{P}_{t+k-1} - \dots - \tilde{P}_{t+1} + \tilde{P}_{t+1} - \tilde{P}_t + \tilde{P}_t - \tilde{P}_{t-1}) \\ &= \sum_{k=0}^{\infty} (\theta\beta)^k \left[(\tilde{P}_{t+k} - \tilde{P}_{t+k-1}) + (\tilde{P}_{t+k-1} - \tilde{P}_{t+k-2}) + \dots + (\tilde{P}_t - \tilde{P}_{t-1}) \right] \\ &= (\tilde{P}_t - \tilde{P}_{t-1}) + (\theta\beta) \left[(\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \\ & \quad + (\theta\beta)^2 \left[(\tilde{P}_{t+2} - \tilde{P}_{t+1}) + (\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \\ & \quad + (\theta\beta)^3 \left[(\tilde{P}_{t+3} - \tilde{P}_{t+2}) + (\tilde{P}_{t+2} - \tilde{P}_{t+1}) + (\tilde{P}_{t+1} - \tilde{P}_t) + (\tilde{P}_t - \tilde{P}_{t-1}) \right] + \dots \end{aligned}$$

$$\begin{aligned}
&= \left(\tilde{P}_t - \tilde{P}_{t-1} \right) \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \\
&\quad + \left(\tilde{P}_{t+1} - \tilde{P}_t \right) \theta\beta \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \\
&\quad + \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) (\theta\beta)^2 \left[1 + \theta\beta + (\theta\beta)^2 + (\theta\beta)^3 + \dots \right] + \dots \\
&= \left(\tilde{P}_t - \tilde{P}_{t-1} \right) \frac{1}{1-\theta\beta} + \left(\tilde{P}_{t+1} - \tilde{P}_t \right) \frac{\theta\beta}{1-\theta\beta} + \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) \frac{(\theta\beta)^2}{1-\theta\beta} + \dots \\
&= \frac{1}{1-\theta\beta} \left[\left(\tilde{P}_t - \tilde{P}_{t-1} \right) + \theta\beta \left(\tilde{P}_{t+1} - \tilde{P}_t \right) + (\theta\beta)^2 \left(\tilde{P}_{t+2} - \tilde{P}_{t+1} \right) + \dots \right] \\
&= \frac{1}{1-\theta\beta} \sum_{k=0}^{\infty} (\theta\beta)^k (\tilde{P}_{t+k} - \tilde{P}_{t+k-1})
\end{aligned}$$

■